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Advances in Hazard Management for Machine Learning based Safety-Critical Systems

Operating in Unpredictable Environments

by

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Thesis for the degree of Doctor of Philosophy

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Abstract

Faculty of Engineering and Physical Sciences

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Appendix G Assessment of Existing Safety Processes

by

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This appendix accompanies the doctoral thesis entitled *Advances in Hazard Management for Machine Learning based Safety-Critical Systems Operating in Unpredictable Environments*. Appendix G presents the full empirical assessment of three established safety analysis methods (HAZOP, STAMP/STPA, and FRAM) applied to the unsafe train tracks case study introduced in §1.4.1. The assessment evaluates each method's capacity for novel hazard discovery and Black Swan scenario prediction under Compounded Unpredictability and Epistemic Uncertainty (CUPE) conditions, providing the evidential basis for the comparative findings summarised in §2.3 and §6.9.

Each method is applied in full to the same problem, following its standard procedure: HAZOP through guide-word-driven deviation analysis (§G.1), STAMP/STPA through hierarchical control structure modelling and unsafe control action identification across four analytical phases (§G.2), and FRAM through functional characterisation, output variability assessment, and functional resonance identification (§G.3). For each method, the resulting predicted situations are recorded and their Analytical Attention Distribution (ATD) is plotted and evaluated against the three ATD modes defined in §3.5.1 (Normal, Uniform, and Long-tailed) to characterise the method-induced attention bias each approach produces.

A Novelty Test (§4.10) is applied to the predicted situations from each method to classify outputs as original or non-original and typical or atypical, establishing a quantitative novelty rate for each approach (§G.5). An AIC-based gap analysis of FRAM (§G.4) further identifies interaction categories and potential system requirements that FRAM's functional ontology structurally excludes, illustrating the coverage limitations that arise from method-bounded attention allocation. The appendix concludes with a consolidated Predicted Situations Distribution Analysis (§G.5), including frequency extraction, statistical property evaluation, and the Python implementation used to compute ATD metrics and generate histogram visualisations.

The findings provide the comparative empirical foundation for the claim that existing safety methods exhibit Normal ATD patterns under CUPE conditions, producing structured coverage blindness in the low-salience tail where Black Swan scenarios reside.

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Research Thesis: Declaration of Authorship

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Title of thesis: Advances in Hazard Management for Machine Learning based Safety-Critical Systems
Operating in unpredictable environments

I declare that this thesis and the work presented in it are my own and has been generated by me as the result of my own original research.

I confirm that:

This work was done wholly or mainly while in candidature for a research degree at this University. Furthermore,

Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated. Furthermore,

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Appendix G Assessment of Existing Safety Processes

G.0 Background

HAZOP, STAMP, and FRAM are methodologies that aim to guide the architect's predictive thinking process (see Section 4.2) to identify potential deviations that may have been overlooked during system architecting. Those methods practically influence how an architect manages their predictive attention on different types of deviations. For example, in HAZOP, the approach prompts the architect to focus on deviations that carry the connotations of “no”, “early”, or “late”. STAMP instructs the architect, in the first step, to define relevant situations and exclude others to specify the boundary of the system of interest. These guidelines serve as thought policies, ultimately influencing the way an architect perceives the complexity of a problem space. Here, missing requirements and misconceptions can lead to an ineffective perception of the problem.

For example, if the problem space's aleatoric uncertainty is characterised by heavy-tailed complexity, that means Black Swan events dominate. If the approach prompts the architect to construct a mental model that oversimplifies reality and treats it as if it is normally distributed complexity, it fundamentally means the architect will miss Black Swan events. Open, complicated environment complexity is not the same as the complexity of a highly controlled situation run by deterministic automated robots, where everything is predictable. In this Appendix, we will attempt to assess HAZOP, STAMP and FRAM systems thinking approach and how they impact the architect's predictive thinking process when dealing with an open, complicated problem.

So, how do we do that, and how can we test the impact of the thinking process on an architect's predictive thinking?

We will examine how HAZOP, STAMP, and FRAM impact our predictive thinking by analysing how these processes influence our predictive attention distribution, using their ontologies and thinking policies. We will assess this by collecting the predicted situations at the most influential step and then analysing their distribution. We will also examine how biased our attention was concerning the ontology they used. For example, HAZOP provides 14 different types of deviations to consider. We regard those types as predictive guiding words because they prompt our predictive attention to imagine what might happen if an observation obeys a particular type of deviation rule.

This Appendix evaluates established safety analysis approaches (HAZOP, STAMP/STPA, and FRAM) using a multi-pronged approach:

- **Primary evidence:** research literature analysis. We review peer-reviewed studies and surveys describing practical applications and reported limitations of HAZOP, STAMP/STPA, and FRAM, focusing on (i) analyst/team dependence, (ii) scalability and effort, (iii) ability to handle socio-technical complexity and learning-enabled/autonomous behaviour, and (iv) validation and reproducibility concerns.
- **Secondary evidence:** Applying case studies to assess each approach. The unsafe train tracks case study (detailed analysis in Appendix G) is used as a complementary illustration to ground the discussion and to investigate whether additional limitations beyond those reported in the literature manifest in a representative autonomous-systems engineering scenario (for open complicated environments). The case study is not used as the primary basis for general claims, since these methods are typically applied by multidisciplinary expert teams and require domain experience.

This positioning reduces threats to validity by ensuring that the critique does not depend solely on a single analyst's application of the methods.

Furthermore, Appendix G showcases our application of these approaches to solve the unsafe train tracks problem, so we will not repeat the work in this Chapter. Instead, we will discuss our experience with these approaches to set a reference for comparison with our approach. The reader is encouraged to review Appendix G for reference.

Kritzinger emphasised that the main problem in safety engineering is not the identified hazards but rather the “unidentified hazards”. **Non-typical-saliency** of a possible hazard can be a distinctive property of hard-to-identify hazards for experts. Hence, we proposed the concept of Non-typical-saliency as an assessment criterion to determine whether an approach is sufficiently equipped to predict them when applied to complex pen environments.

This provides a suitable criterion for comparing the ability of safety approaches to identify non-typically-salient hazards in an open, complicated environment. After all, in Section 4.1, we explained the nature of such problems, which are often dominated by non-typically-salient hazards and Black Swans. Ensuring autonomous systems perform as expected during Black Swan operations is critical to their safety. Therefore, we chose a non-typically-salient hazard and assessed legacy APPROACHES (HAZOP, STAMP, FRAM) to bring us closer to discovering the non-typically-salient hazard. We expect legacy approaches' control-theory paradigm renders them insufficient to predict non-typically-salient and emergent hazards associated with the complexity of Compounded Epistemic Uncertainty problems.

Traditional systems engineering approaches to date can be classified as a pure control-theory-based approach to complexity. Meaning, the Architect assumes that any complexity field can be viewed like an electronic board where we have specifiable components with clear, unchanging

boundaries. While such simplification proved to be useful for traditional deterministic cps problems, it may not be effective worldview of open, complicated environments and autonomous systems. We experience this attitude towards complexity first-hand in this Chapter as we examine their application to solve our proposed problem.

The main problem we are trying to address in our approach can be described in two dimensions:

- Addressing the problem of predicting harder-to-predict, non-typically-salient, unforeseen emergent scenarios related to unexpected emergent behaviour of autonomous systems in harder-to-predict complicated operational environments.
- Ascertaining that emergent behaviours fulfil requirements (especially safety ones) are preserved when autonomous systems interact with their environment in any possible emergent scenario that may or may not have been thought of during design and training time.

Both dimensions are addressed in our approach. Our solution for handling complicated emergence problems is to enhance the Architect's predictive cognition through the use of systemic (through the application of general systems intuition) and systematic thinking approaches. The second concern is that we propose justification-based traceability of how artefacts are developed, coupled with the demonstration of the thinking and intuition employed in their development through the application of STCoT approaches. Each stage of our approach prompts the Architect to address the initial problem from a different perspective. Imagine a problematic situation, such as an unknown alien object landing in front of the Architect, who has never encountered such a level of complexity.

Since the object has never been seen before, the Architect's predictive thinking ability experiences a state of confusion. Our approach aims to provide the Architect with generic systems-thinking approaches that can guide and streamline their application regardless of how unknown and alien a problematic situation is. We view the power of adapting general systems theories as making systems engineering approaches problem-agnostic. Attempting to predict the future complexity and complicatedness of an unknown complicated problem is an example of a confusing complex.

The following is the problem brief:

An anonymous drone intruded into a bounded train track zone while the train passed by. Police officers were called to the scene and tried to capture the drone but were unsuccessful.

G.0.1 Thought experiment: assessment question

Before we start the approach, we would like to prompt your predictive thinking about the scenario. We have a particular reason for doing so, which will be revealed later. Let's take you back to the case study we described in Section 1.4.1. Imagine a situation like in the image below:



Figure G.1 A snapshot taken from DuckDuckGo web browser maps of a strip of train tracks in Holy Town, Scotland, UK

Based on personal engineering judgement alone, what would be your answer for the following enquiries about the problem complexity? Answer and record your answer for the following questions:

1. Based on images alone, is the track zone completely unprotected against low-altitude drone entry?
2. Could trackside vegetation and its interaction with the track zone plausibly further assist adversarial drone intrusion, thereby affecting the safety of the track zone?
3. Can the oscillation of a tree branch due to wind or train motion turbulence lead to an unsafe situation?

We will describe the approaches and then demonstrate their application in selected examples.

In this Chapter, we will apply three approaches to 2 aspects to compare the results of our approach with theirs. We do not intend to apply the approaches comprehensively; rather, we will apply only the main concepts of each approach. The idea is that we want to compare how each approach handles two systems engineering problems:

- Articulating the problem domain to discover problems that were not considered initially.
- Discover Black Swan events and associated hazards. Especially related to computer vision anomalies.

G.0.2 Novelty Test for predicted situations

Novelty Test assesses whether a predicted knowledge (such as a situation or a hazard):

- **Originality:** The originality of the predicted situation with respect to some lexicon of analysis of prior problems (from existing literature, or other case studies). Quantified by including the predicted situations in a different case study (one or more) and examining whether they have been mentioned in other analysis literature. The originality score is computed by the inverse of frequency.
- **Typicality Assessment:** Assess whether a predicted situation would (or would not) be readily (typically) identified by a competent, expert. The qualification would be either typical or atypical (not typical).

Specifically, the expert determines whether it lies (or doesn't) outside standard component-level, deterministic cause-and-effect framing, is counterintuitive or "out of scope." A situation fails to be atypically salient, test if it is "Obvious" (routine, checklistable, or immediately mappable to known categories) and passes only if it is emergent or perspective-shifting rather than an expected failure mode.

The Novelty Test procedure involves **(a)** conducting a thinking approach (implementing a thought step), **(b)** identifying predicted situations according to some convention, for example, in format of {adjective_noun} as and when relatively new knowledge discovered, **(c)** extracting predicted situations and classify them according to some ontology (situations types, guideword, deviations, etc.), **(d)** compute a tally of mentions, **(e)** compute attention score based on probability distribution of frequency to total of predicted situation, **(f)** Determine originality by measuring the frequency of predicted situations relative to a prior literature such risk assessments or repository of possible failures modes, (which literature to choose for assessing general originality is subject to expert judgement) **(e)** generate a histogram and distribution fit to evaluate the skewness and kurtosis of the predicted situations thus revealing the saliency of situations to the analyst. For a detailed description of the approach taken in the proposed thesis, see §3.13, G.5.

a) Typicality Test for novelty of predicted hazards

An essential quality of safety and security tools (and systems engineering approaches) is their ability to identify novel hazards Kritzinger [69] emphasised that the main problem is not the easily identifiable hazards alone but the "unidentified hazards" too. Once a hazard is identified, the Architect can plan its mitigation; unknown hazards are surprising no design team wants to encounter. Based on this motivation, the following rubric is designed to evaluate the predictive potential of safety approaches in predicting novel hazards. The proposed approach does not claim a complete, infallible formalisation, but rather a general qualitative guide for determining whether a hazard is novel. To that extent, the proposed approach aims to establish a body of work that categorises predictive thinking (see §3.6) to help evaluate the thinking approaches. Deeper formalisation will be considered as part of future work.

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This is a rubric-based test to determine whether a predicted situation, using a thinking approach, is typical or non-typical (i.e. atypical/novel). The central assumption made for this type of philosophical test is:

To effectively address the compounded epistemic uncertainty problem, the systems engineering approach must enable a novice architect to predict non-typically-salient failure modes (Black Swans).

A key criterion for evaluating legacy approaches is whether they enable us to identify non-typically salient hazards that a skilled, objective reality only framing safety engineer cannot develop using the approach. Judging the non-typical-saliency of a predicted hazard can benefit from the following qualifications:

- **Typical:** Not outside the expected relevance in such a scenario or problem space/
Atypical: Outside expected relevance in such a scenario or problem space.
- **Typical:** High prior risk assessment is available/**Atypical:** Low or no prior risk assessment is available.
- **Typical:** Typically, notable in this type of problem space (operational domain)/**Atypical:** Not typically notable in this type of problem space (operational domain).
- **Typical:** It can be objectively quantified. If there is a risk, the risk can be quantified and can be reasonably guaranteed/**Atypical:** it cannot be objectively quantified, if there is a risk, the risk cannot be quantified, nor can be reasonably guaranteed.
- **Typical:** Its impact is not open for interpretation by ML or Analyst/ **Atypical:** Its impact is open for interpretation by ML or Analyst.

Note 1: The criteria are developed to help a practitioner assess a predicted situation during analysis. Both criteria describe the same notion, “control-framing predicts relatively typical events,” in slightly different ways. The test is considered a rubric to facilitate understanding.

Note 2: see §4.10 for a more concrete definition of the Novelty Test.

b) Novelty Test Table

The analysis output was then presented in Table G.0.2 (for more detail see section H.4.4.31), where each potential contributing scenario (referred to as a predicted situation) was qualitatively evaluated against the above criteria. The following is an enumeration of how to conduct the test:

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Table G.0.2 Example of Novelty Test for Case Study 1: Stage 1 approach

Predicted Situation	Tally	Attention%	Originality	Typicality Test	Rational for typicality value	Novelty Test
fallen vegetation over the train tracks structure	7	1.71	original	Typical	Not outside the expected relevance in such a scenario or problem space	Not novel
police officer situational awareness	6	1.46	original	Atypical	It cannot be objectively quantified. If there is a risk, the risk cannot be quantified, nor can it be reasonably guaranteed	Novel

The following definitions describe Table G.0.2:

- Predicted Situation:** captures situation label (e.g., train_derailment, roaming_adversarial_drone). These labels represent distinct contributing situations that the analysis surfaced.
- Tally:** how many times the situation was referenced during analysis. This is treated as an empirical proxy for *allocated predictive attention*: repeated mentions indicate that the method and analyst returned to that situation, refined it, or used it repeatedly to reason about downstream effects.
- Originality (Original/Not original):** how original the predicted situation is relative to prior literature about the problem space. It could be a repository of typical failures, or a project team-selected set of previous studies on the topic of safety. The prior literature can be any corpus of information to compare to. In this thesis, the predicted situations in Appendix G were used as prior literature to compare the outputs of the thinking approaches.
- Typicality Test (Typical/Atypical):** an expert-driven saliency classification outcome for that predicted situation. With respect to objective-reality framing, a predicted situation or scenario is classed as “Typical” if it can be objectively quantified, its risk can be defended, and it has substantive prior literature or is expected within the problem space across different scenarios. Being typical also means its impact is not open to different interpretations by the analyst. **Atypical:** Novel and specific to this particular problem, can not possibly be encountered in any other problematic scenarios related to railway safety and reliable operation. It cannot be objectively quantified, nor can its risk. Its impact is subject to interpretation by the ML and Analyst.
- Rational for typicality value:** describe an explanation that justifies the chosen saliency values. The project may provide a detailed description to justify the rationale behind the chosen typicality value, or use the following generic criterion:
 - Not outside the expected relevance in such a scenario or problem space/Outside expected relevance in such a scenario or problem space.
 - High prior risk assessment is available/Low or no prior risk assessment is available.
 - Typically, notable in this type of problem space (operational domain)/Not typically notable in this type of problem space (operational domain).

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- It can be objectively quantified. If there is a risk, the risk can be quantified and can be reasonably guaranteed/it cannot be objectively quantified, if there is a risk, the risk cannot be quantified, nor can be reasonably guaranteed.
- Its impact is open for interpretation by ML or Analyst/Its impact is not open for interpretation by ML or Analyst

Note that the Novelty Test criteria are only guidelines. They all refer to the same concept, whether a prediction is novel or not. Either option can be used. For example, one value would be “Not counterintuitive or outside the expected relevance”, or just “not typically notable”. This is because qualitatively judging whether a situation or hazard is novel can be ambiguous.

An example of a typically salient hazard would be;

- An adversarial drone that enters the train track earlier than expected.

If the approach made you notice the obvious thing for example an adversarial drone entering the airspace, nothing new about it. This is not something surprising given the main problem is that. Another set of obvious failures or easy to spot without the help of the approach:

- Adversarial drone not operational: leads to falling from the sky.
- Adversarial drone operational but hit the electrical power lines or infrastructure.

An example of a non-typically-salient hazard would be;

- A flying balloon with an image of a drone on it leads the Eagle Drone to think it is an actual adversarial drone. Imagine many balloons being blown onto the train track.
- A laser with a drone-shaped stencil spoofing the eagle drone to think there is a drone on the wall flying straight into it, perhaps the train itself.
- A new gloss paint on the train that startled an adversarial drone from recognising it as a train.
- Another example of a non-typically-salient hazard: a Tesla Model perception system confuses a moon with a traffic light.

G.1 Applying HAZOP method

This process will attempt to apply HAZOP methods to solve two aspects. The first is articulating the problem, and the other is articulating the hazards associated with introducing the Eagle Drone to counter roaming_adversarial_drone behaviour.

G.1.1 Problem articulation

In this section, we will be applying the HAZOP method for problem articulation prior to the introduction of the Eagle Drone. The HAZOP study requires an existing design and system schematics. In this instance, we will model the problem domain and use this as a schema to apply the HAZOP analysis. To initiate the articulation, we require a schematic that describes the system. The system itself is the problem. We will use a generic block diagram to model the interactions in the problem domain (See G.1):

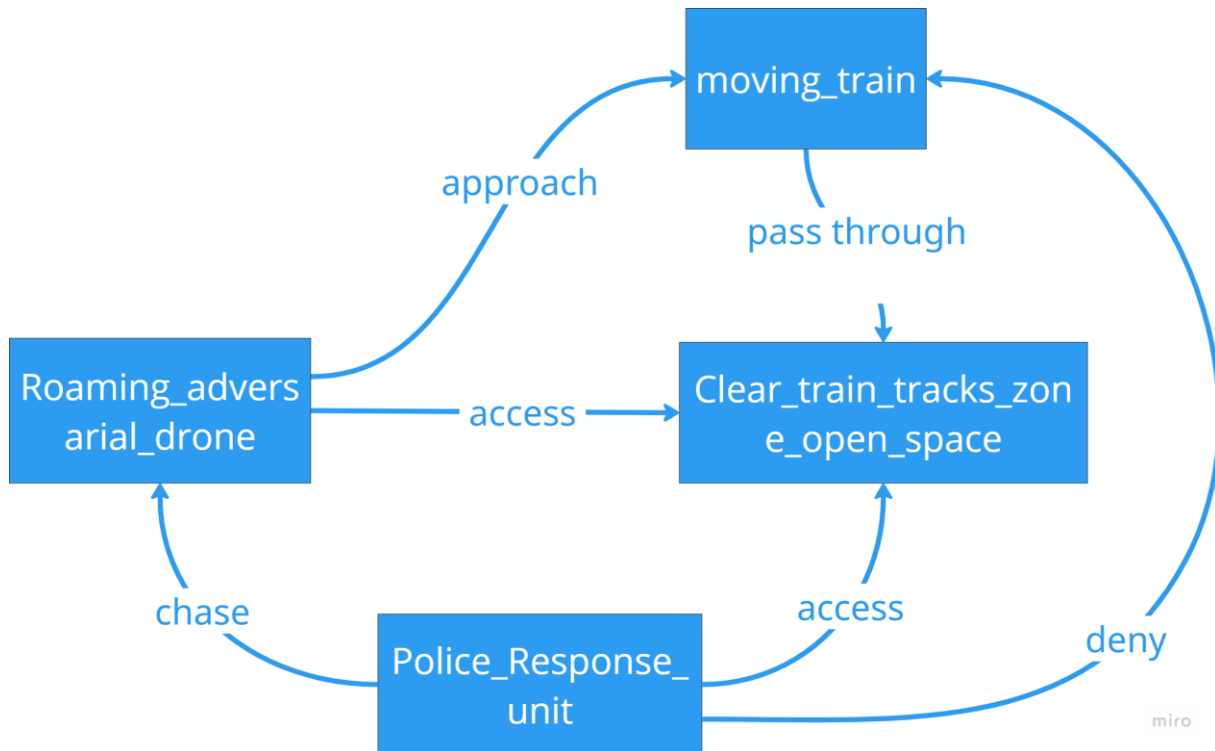


Figure G.2 Control-based schema for HAZOP study

A system diagram illustrating interactions between different situations: a **moving_train**, a **clear_train_tracks_zone_open_space**, a **roaming_adversarial_drone**, a **Police_Response_unit**, and the relationships (or interactions) between them. Below is a list of all the possible interactions based on the diagram, which can be used for a HAZOP (hazard and operability) study. The following list of interactions were used deviation analysis:

1. {moving_train} pass through {clear_train_tracks_zone_open_space}: the moving_train passes through the clear_train_tracks_zone_open_space.
2. {roaming_adversarial_drone} access {clear_train_tracks_zone_open_space}: the roaming_adversarial_drone accesses the clear_train_tracks_zone_open_space.
3. {police_response_unit} chase {roaming_adversarial_drone}: the Police_Response_unit chases the roaming_adversarial_drone.
4. {police_response_unit} access {clear_train_tracks_zone_open_space}: the Police_Response_unit gains access to the clear_train_tracks_zone_open_space.
5. {roaming_adversarial_drone} – {moving_train}: the roaming_adversarial_drone approaches the train.

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6. {police_response_unit} deny {moving_train}: the Police_Response_unit denies moving_train access.

To make HAZOP work as a problem articulation method, we will need to reinterpret the guidewords:

Table G.1 HAZOP guidewords meaning

Guideword	Adapted meaning
No	None of the individual systems achieve their desired intent.
More	Quantitative increase in a parameter
Less	Quantitative decrease in a parameter
As well as	An additional activity occurs.
Part of	Individual systems partially achieve the desired intention.
Reverse	The logical opposite of the individual systems design intention occurs.
Other than	Complete substitution occurs, or an unusual condition exists.
Where else	Applicable for flows, transfers, sources, and destinations
Before/after	Step (or part of it) is affected out of sequence.
Early/late	Timing is different from the intention.
Faster/slower	The step is done/not done with the right timing.

Next, we create the Table that captures the HAZOP process:

Table G.2 HAZOP process from problem articulation

Interaction	Guideword	Cause [problem articulation step]	Consequence [problem articulation step]	Safeguards [solution step]	Actions [solution step]
moving_train - Clear_train_tracks_zone_open_space	No	Obstruction on the track {train_tracks_obstruction}, equipment failure {equipment_failure}, and signalling malfunction {signalling_malfunction}	Train delay {train_delay}, potential derailment {train_derailment} if an object is hit	Surveillance cameras, track inspection {available_ground_based_security_systems} protocols	Ensure regular inspection and provide real-time obstruction alerts
moving_train - Clear_train_tracks_zone_open_space	Late	Delayed or advanced scheduling {train_delay}	Increased risk of collision or hazards if other operations are unprepared {all_trains_operations}	Automated scheduling and signalling systems {train_delay}	Additional scheduling checks, implement manual override

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					protocols
Roaming_adversarial_drone – Clear_train_tracks_zone_open_space	No	No deterrence by the track zone {clear_train_tracks_zone_open_space}	Potential collision with a train {adversarial_drone_collision_with_train}	Missing safeguards	Increase enforcement measures and add automated drone interception mechanisms.
Roaming_adversarial_drone – Clear_train_tracks_zone_open_space	More	Increased roaming_adversarial_drone activity {more_roaming_adversarial_drones}	Higher likelihood of a train-drone collision {adversarial_drone_collision_with_train}, all_trains_operations disruption {all_trains_operations}	Missing safeguards	Implement stricter airspace controls and drone deterrence systems
Police_Response_unit – roaming_adversarial_drone	Faster	Faster, inaccurate assessment of the drone's risk level {roaming_adversarial_drone}	Officers may enter moving_train zone without precautions {unsafe_police_officer_movements}, leading to safety risks.	Missing safeguards	Define threat levels and response protocols
Police_Response_unit – roaming_adversarial_drone	Other than	Misidentification of the roaming_adversarial_drone {roaming_adversarial_drone}	Resources misallocated, delay in response to actual threat {adversarial_drone_response_strategies}	Missing safeguards	Enhance identification accuracy with ai-assisted drone recognition
Police_Response_unit – clear_train_tracks_zone_open_space	As well as	Miscommunication about moving_train timing or lack of situational awareness {adversarial_drone_response_strategies}	High risk of collision {train_derailment}, endangering officer {unsafe_police_officer_movements}	Entry control systems, real-time tracking of officers	Implement alert systems, define restricted access times
Police_Response_unit – clear_train_tracks_zone_open_space	Early	Communication errors {adversarial_drone_response_strategies}, {police_comms_errors}	Extended exposure to potential hazards in track zone {unsafe_train_tracks_zone_structure}	Access coordination, timing protocols	Review and improve timing protocols for entry
roaming_adversarial_drone – moving_train	Reverse	Drone control malfunction {roaming_adversarial_drone}	Potential collision, damage to moving_train {train_derailment} infrastructure {unsafe_train_tracks_zone_structure}, passenger risk {unsafe_unsecured_p	Missing safeguards	Enable police or automated systems to disable/intercept

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			assengers_and_goods }		intruding drones
roaming_adversarial_drone – train	Faster	Quick speed to ingress the zone {roaming_adversarial_drones_location_data}	Increased likelihood of collision {train_derailment}, reduced response time for mitigation {adversarial_drone_response_strategies}	Missing safeguards	Implement rapid drone interception measures
Police_Response_unit – moving_train	No	Restricted access, malfunction of access points {unsafe_police_officer_movements}	Inability to mitigate threat or access moving_train for safety {police_officer_capture_adversarial_drone}	Emergency access protocols, clear entry points	Install alternate entry systems, and enhance emergency accessibility
Police_Response_unit – train	Less	Inaccurate timing {inaccurate_response_timing}, restricted access {unsafe_police_officer_movements}	Insufficient time to respond to threats or ensure area safety {inaccurate_response_timing},	Communication systems, access timing controls	Improve timing and access protocols to ensure officer safety

For the same interaction, the following table captures the keywords which regarded as irrelevant:

Table G.2A Guide word coverage per interaction

Interaction	No	More	Less	As well as	Part of	Reverse	Other than
1	used	0	0	0	0	0	0
2	used	used	0	0	0	0	0
3	0	0	0	0	0	0	used
4	0	0	0	used	0	0	0
5	0	0	0	0	0	used	0
6	used	0	used	0	0	0	0
Interaction	Where else	Before	after	Early	late	Faster	slower
1	0	0	0	0	used	0	0
2	0	0	0	0	0	0	0
3	0	0	0	0	0	used	0
4	0	0	0	used	0	0	0
5	0	0	0	0	0	used	0
6	0	0	0	0	0	0	0

From which the following types of possible deviations were considered as irrelevant, thus received no attention:

Table G.2B Missed types of deviation due to being considered as out-of-scope or irrelevant

Discarded interaction	Attention
-----------------------	-----------

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n/a_More_interaction_1	0
n/a_Less_interaction_1	0
n/a_As well as_interaction_1	0
n/a_Part of_interaction_1	0
n/a_Reverse_interaction_1	0
n/a_Other than_interaction_1	0
n/a_Where else_interaction_1	0
n/a_Before_interaction_1	0
n/a_after_interaction_1	0
n/a_Early_interaction_1	0
n/a_Faster_interaction_1	0
n/a_slower_interaction_1	0
n/a_Less_interaction_2	0
n/a_As well as_interaction_2	0
n/a_Part of_interaction_2	0
n/a_Reverse_interaction_2	0
n/a_Other than_interaction_2	0
n/a_Where else_interaction_2	0
n/a_Before_interaction_2	0
n/a_after_interaction_2	0
n/a_Early_interaction_2	0
n/a_late_interaction_2	0
n/a_Faster_interaction_2	0
n/a_slower_interaction_2	0
n/a_No_interaction_3	0
n/a_More_interaction_3	0
n/a_Less_interaction_3	0
n/a_As well as_interaction_3	0
n/a_Part of_interaction_3	0
n/a_Reverse_interaction_3	0
n/a_Where else_interaction_3	0
n/a_Before_interaction_3	0
n/a_after_interaction_3	0
n/a_Early_interaction_3	0
n/a_late_interaction_3	0
n/a_slower_interaction_3	0
n/a_No_interaction_4	0
n/a_More_interaction_4	0
n/a_Less_interaction_4	0
n/a_Part of_interaction_4	0
n/a_Reverse_interaction_4	0

n/a_Other than_interaction_4	0
n/a_Where else_interaction_4	0
n/a_Before_interaction_4	0
n/a_after_interaction_4	0
n/a_late_interaction_4	0
n/a_Faster_interaction_4	0
n/a_slower_interaction_4	0
n/a_No_interaction_5	0
n/a_More_interaction_5	0
n/a_Less_interaction_5	0
n/a_As well as_interaction_5	0
n/a_Part of_interaction_5	0
n/a_Other than_interaction_5	0
n/a_Where else_interaction_5	0
n/a_Before_interaction_5	0
n/a_after_interaction_5	0
n/a_Early_interaction_5	0
n/a_late_interaction_5	0
n/a_slower_interaction_5	0
n/a_More_interaction_6	0
n/a_As well as_interaction_6	0
n/a_Part of_interaction_6	0
n/a_Reverse_interaction_6	0
n/a_Other than_interaction_6	0
n/a_Where else_interaction_6	0
n/a_Before_interaction_6	0
n/a_after_interaction_6	0
n/a_Early_interaction_6	0
n/a_late_interaction_6	0
n/a_Faster_interaction_6	0
n/a_slower_interaction_6	0

G.1.2 Predictive Attention Distribution as guided by HAZOP

Let's look at the predictive thinking output in terms of predicted situations in Table G.3. We notice that our attentive mentions count of situations (within the problem analysis sections for Table G.3, which are columns “Cause”, “Consequence” only not the solution), we notice the following predictive behaviour:

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Table G.3 Novelty Test on the predicted situations using HAZOP method

Predicted situation	Tally	Arch. Attention%	Originality	Typicality Test	Rational for typicality value	Novelty Test
roaming adversarial drone	7	14.29	not original	Atypical	Not typically notable in this type of problem space (operational domain)	Not novel
uncertain clear train tracks zone open space	4	8.16	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
likely derailed train	4	8.16	not original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
unsafe police officer movements	4	8.16	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
successful adversarial drone response strategies	4	8.16	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
unsafe moving train	3	6.12	not original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
uncertain police response unit	3	6.12	not original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
frequently delayed train	3	6.12	not original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
impacted all trains operations	2	4.08	not original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
highly probable adversarial drone collision with unsafe moving train	2	4.08	not original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
unsafe train tracks zone structure	2	4.08	not original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
likely inaccurate response time	2	4.08	not original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
uncertain police response	1	2.04	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
visible train tracks obstruction	1	2.04	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
failed equipment	1	2.04	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
signalling malfunction	1	2.04	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
available ground-based security systems	1	2.04	not original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
more roaming adversarial drones	1	2.04	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
likely police comms errors	1	2.04	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
unsafe unsecured passengers and goods	1	2.04	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
police officer capture adversarial drone	1	2.04	original	typical	Not outside the expected relevance in such scenario or problem space	Not novel
n/a More interaction 1	0	negligible				
n/a Less interaction 1	0	negligible				
n/a As well as interaction 1	0	negligible				
n/a Part of interaction 1	0	negligible				
n/a Reverse interaction 1	0	negligible				
n/a Other than interaction 1	0	negligible				
n/a Where else interaction 1	0	negligible				
n/a Before interaction 1	0	negligible				
n/a after interaction 1	0	negligible				
n/a Early interaction 1	0	negligible				
n/a Faster interaction 1	0	negligible				
n/a slower interaction 1	0	negligible				
n/a Less interaction 2	0	negligible				
n/a As well as interaction 2	0	negligible				

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n/a Part of interaction 2	0	negligible				
n/a Reverse interaction 2	0	negligible				
n/a Other than interaction 2	0	negligible				
n/a Where else interaction 2	0	negligible				
n/a Before interaction 2	0	negligible				
n/a after interaction 2	0	negligible				
n/a Early interaction 2	0	negligible				
n/a late interaction 2	0	negligible				
n/a Faster interaction 2	0	negligible				
n/a slower interaction 2	0	negligible				
n/a No interaction 3	0	negligible				
n/a More interaction 3	0	negligible				
n/a Less interaction 3	0	negligible				
n/a As well as interaction 3	0	negligible				
n/a Part of interaction 3	0	negligible				
n/a Reverse interaction 3	0	negligible				
n/a Where else interaction 3	0	negligible				
n/a Before interaction 3	0	negligible				
n/a after interaction 3	0	negligible				
n/a Early interaction 3	0	negligible				
n/a late interaction 3	0	negligible				
n/a slower interaction 3	0	negligible				
n/a No interaction 4	0	negligible				
n/a More interaction 4	0	negligible				
n/a Less interaction 4	0	negligible				
n/a Part of interaction 4	0	negligible				
n/a Reverse interaction 4	0	negligible				
n/a Other than interaction 4	0	negligible				
n/a Where else interaction 4	0	negligible				
n/a Before interaction 4	0	negligible				
n/a after interaction 4	0	negligible				
n/a late interaction 4	0	negligible				
n/a Faster interaction 4	0	negligible				
n/a slower interaction 4	0	negligible				
n/a No interaction 5	0	negligible				
n/a More interaction 5	0	negligible				
n/a Less interaction 5	0	negligible				
n/a As well as interaction 5	0	negligible				
n/a Part of interaction 5	0	negligible				
n/a Other than interaction 5	0	negligible				
n/a Where else interaction 5	0	negligible				
n/a Before interaction 5	0	negligible				
n/a after interaction 5	0	negligible				
n/a Early interaction 5	0	negligible				
n/a late interaction 5	0	negligible				

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n/a slower interaction 5	0	negligible				
n/a More interaction 6	0	negligible				
n/a As well as interaction 6	0	negligible				
n/a Part of interaction 6	0	negligible				
n/a Reverse interaction 6	0	negligible				
n/a Other than interaction 6	0	negligible				
n/a Where else interaction 6	0	negligible				
n/a Before interaction 6	0	negligible				
n/a after interaction 6	0	negligible				
n/a Early interaction 6	0	negligible				
n/a late interaction 6	0	negligible				
n/a Faster interaction 6	0	negligible				
n/a slower interaction 6	0	negligible				

Table G.3 summarises the outputs produced when HAZOP was applied to the scenario and then evaluated using the Typicality Test. Each row represents a predicted situation label extracted from the analysis record (using the label-persistence convention), along with three interpretive measures.

- **Predicted situation:** captures situation label (e.g., train_derailment, roaming_adversarial_drone). These labels represent distinct contributing situations that the analysis surfaced.
- **Number of attentive mentions:** how many times the situation was referenced during analysis. This is treated as an empirical proxy for *allocated predictive attention*: repeated mentions indicate that the method and analyst returned to that situation, refined it, or used it repeatedly to reason about downstream effects.
- **Typicality Test:** the saliency classification outcome for that predicted situation. “Typical” if comply with Typicality Test criteria, otherwise, “atypical”. It also include mentions the criterion that satisfies:
 - Not counterintuitive /Counterintuitive.
 - Not outside the expected relevance / Within expected relevance.
 - High prior literature available / Low or no prior literature.
 - Typically notable / Not typically notable.
- **Rational for saliency value:** describe an explanation that justifies the chosen saliency values.

Note that the Typicality Test criteria are only guidelines. They all refer to the same concept, whether a prediction is novel or not. Either option can be used. For example, one value would be “Not counterintuitive or outside the expected relevance”, or just “not typically notable”. This is because qualitatively judging whether a situation or hazard is novel can be ambiguous.

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Interpreting the table: the entries shown are all marked **typical**, meaning HAZOP's attention, in this excerpt, concentrates on situations that are readily anticipated from the initial scenario framing. For instance:

- train_derailment is a conventional high-consequence rail hazard and commonly appears in routine safety reasoning.
- roaming_adversarial_drone is explicitly present in the scenario premise and therefore highly foregrounded.
- police_response_unit and adversarial_drone_response_strategies are plausible and standard operational considerations once a drone threat is assumed.
- clear_train_tracks_zone_open_space and train_delay represent obvious environmental and operational conditions that are typically explored early in hazard analysis.

As a result, Table G.3 indicates that (at least for the situations listed) HAZOP primarily reinforces baseline, intuitive hazard categories rather than systematically eliciting atypically salient, emergent, or counterintuitive hazards. This does not imply HAZOP is ineffective; rather, it suggests that its contribution (in this portion of the analysis) is strongest in structuring and prioritising already-salient hazards, which aligns with its traditional use in bounded, control-oriented contexts.

If we distribute the number of situations predicted per a guide word, we receive an indication on how the guide word approach had impacted our predictive thinking.

Table G.4 Distribution of predicted situations over HAZOP guide words categories.

Deviation guide word	No. predicted situations
No	9
faster	5
Reverse	4
more	3
As well as	3
Early	3
less	3
late	2
other than	2
Part of	0
Where else	0
Late	0
After	0
Slower	0
yes	0
Intermittent	0

Oscillatory	0
-------------	---

Table G.4 presents the distribution of 34 predicted situations across 17 HAZOP guideword categories reveals a striking imbalance: 8 guidewords generated zero situations, while the remaining attention concentrated heavily on 9 categories, "No" (9 situations), "Faster" (5), "Reverse" (4), and a cluster of moderate-frequency bins ("More," "As well as," "Early," "Less" at 3 each). Notably, the 8 guidewords received zero analytical attention, indicating that these deviation types were either dismissed as irrelevant to the scenario or unconsciously neglected during the HAZOP session.

If we compute statistical properties, such as kurtosis and skewness, they reveal interesting insights regarding how we allocated our analytical attention. The computed statistical properties of this distribution, kurtosis = -0.59 and skewness = -0.04, offer critical insights into the thinking regime induced by the HAZOP methodology. The negative kurtosis (platykurtic, below zero) indicates thin tails and the absence of extreme outliers, consistent with a distribution in which rare or low-salience deviation categories are systematically pruned rather than explored. The near-zero skewness suggests approximate symmetry around the central tendency, yet the presence of multiple zero bins indicates that this symmetry is achieved through truncation: entire guideword categories were excluded from consideration rather than assigned low but non-zero frequencies.

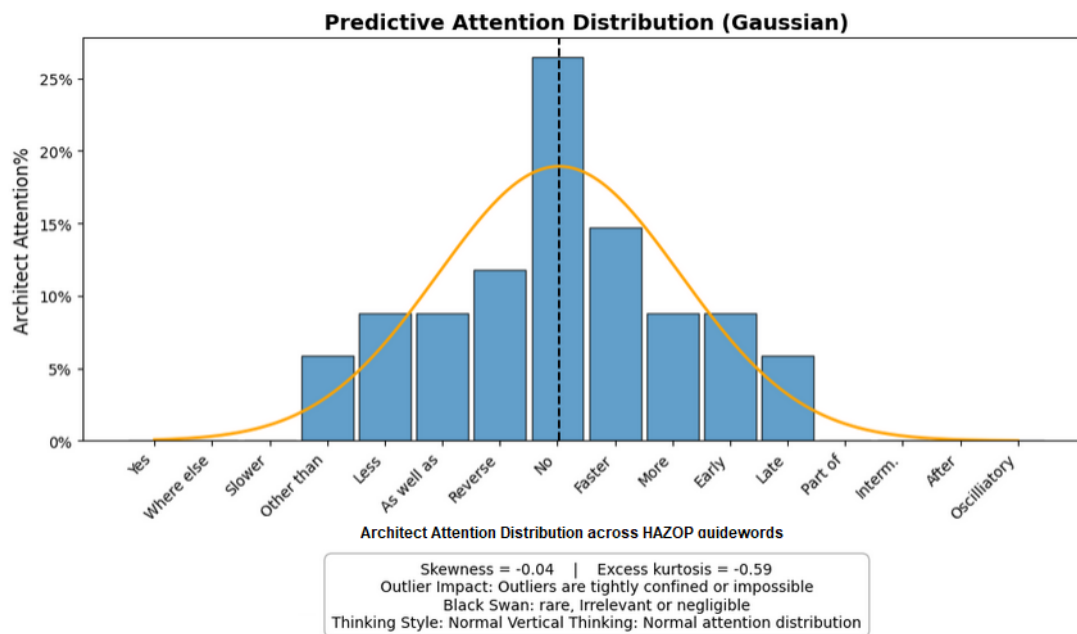
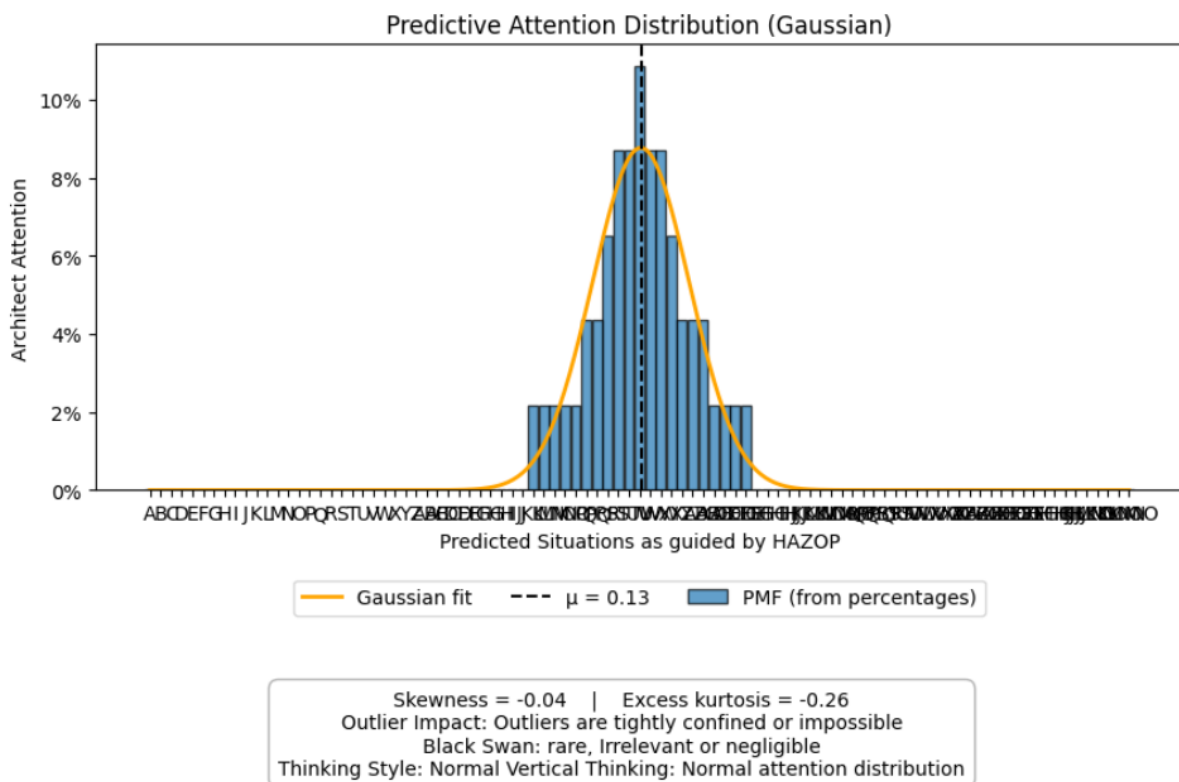


Figure G.3 Impact of HAZOP predictive thinking policies on our predictive attention distribution

The HAZOP process modelling rules prompted us to focus on the types of failures relevant to the observed interactions. Typically, HAZOP commences with a schematic of a tightly controlled process, in which it is evident how each element is directly and precisely coupled, ensuring that one end influences the other in all operability-related cases. However, applying this assumption

to a complex, open environment is invalid. Consequently, it was necessary to develop a schematic to understand the nature of the problem, acknowledging that our simplification already presents an unfaithful representation of the phenomena beyond the obvious elements: an adversarial drone, a police officer, a train, and a track zone. However, we needed something to initiate the process, so we created the model shown in Figure G.1.

In a long-tailed, open, complicated complexity, there is no such determinism, nor is there a solid, unchanging set of possible states of any particular subcomplex. Now, suppose the approach suggests considering only the following types of state-changers: {no, late, early, slower, faster, more}. In that case, any state-change predictors outside this set should not be considered. This means that a potential negation change is neither considered a type worth paying attention to nor predicted as a likely event or interaction. For example, what about a type of state change that is neither “no”, early, late, fast, nor slow? For example, “yes”, “intermittent” and “cyclic” deviation types. They can also be considered types of events or failures that would otherwise have been excluded from HAZOP. See Table G.4 for example, missing failures and thus missing requirements for an open complicated problem.



G.2 Applying STAMP/STPA method

In this session, we will attempt to apply the STAMP methodology to the same aspects as HAZOP and see whether methods help us to arrive to some black swan events. Black swan events are discovered from unexpected or unfamiliar interactions within the analysed complexity field.

G.2.1 Problem articulation

The assumption to be made here for STAMP to work as a problem articulation technique is that the problem controller is the system whose control actions are leading to the undesired complexity. In this case, we will imagine a virtual track zone safety controller (TZSC) a hypothetical control solution whose regulatory actions are inadequate to keep the track zone safe. As such, we can follow the STPA process:

G.2.2 Phase 0: Foundation

c) Establish Safety Objectives

Purpose of the analysis: the objective is to assess and enhance the track zone safety controller (TZSC) in preventing unauthorised drone intrusion into a defined clear_train_tracks_zone_open_space and ensuring the safety of people, infrastructure, and all_trains_operations. The TZSC aims to mitigate losses by controlling access to the track zone's ground and airspace.

G.2.3 Phase 1: System-Level Analysis

a) Step1: Identify Losses & Hazards

Losses (L): Undesirable outcomes (safety, environmental, etc.), ex: L1: Passenger injury

Hazards (H): System states leading to losses, Ex: H1: Vehicle exceeds safe speed in curve

Safety Constraints (SC): System-level rules to prevent hazards, SC1: Speed must adapt to road curvature

(1) Potential losses

1. **L1: Loss of human life or injury {unsafe_unsecured_passengers_and_goods}:** Unauthorised drones intruding {roaming_adversarial_drone} into the track zone may collide with a train {adversarial_drone_collision_with_train}, staff {unsafe_police_officer_movements}, or bystanders {human_intruders}, resulting in injury or fatality.
2. **L2: Damage to property {unsafe_train_tracks_zone_structure}:** intruding drones might damage critical infrastructure, such as the {moving_train} or track components {equipment_failure}, and surrounding structure {outer_tracks_region_hazards}.
3. **L3: Disruption of {all_trains_operations}:** drone intrusion could force trains to halt, reroute, or delay, causing significant disruptions in railway service.
4. **L4: Loss of security:** unauthorised drone presence {roaming_adversarial_drone} increases the risk of security breaches, including unauthorised data access related to sensitive {all_trains_operations}, {track_zone_monitoring_systems_data}.

(2) Identifying system-level hazards

System hazards: the following hazards outline conditions under which the TZSC could fail to ensure track zone safety due to unauthorised drone intrusion:

1. **H1: open airspace {clear_train_tracks_zone_open_space} for unauthorised aerial intrusion {roaming_adversarial_drone}:** The TZSC architecture lacks airspace control capabilities by design, making drone intrusion inherently possible.
2. **H2: failure to trigger alerts {alert_systems} or response actions {police_response_unit} to neutralise intrusion:** TZSC fails to communicate the drone threat {roaming_adversarial_drones_location_data} to {moving_train} control, risking unsafe {all_trains_operations}.
3. **H3: Communication failures between TZSC and {reliable_railways_traffic_control}:** A failure in communication links between the TZSC and the railway control can result in trains moving through the track zone when a drone threat is present, risking collision.
4. **H4: Insufficient response authority for {police_force_existing_capability}:** police_force may lack the technical capacity to neutralise unauthorised drones in the track zone {police_officer_capture_adversarial_drone}. [This hazard refers to organisational constraints, not a technical failure only]

(3) Identifying system-level safety constraints

We can define the constraints by understanding the hazards and their impacts. In the context of problem articulation, constraints refer to solutions or measures to mitigate system-level hazards.

Table G.5 system-level constraints

Hazard	Description [problem articulation step]	Potential impact [problem articulation step]	System-level constraints [solution step]
H1: open airspace for unauthorised aerial intrusion	Drone not detected due to design or system failure {track_zone_monitoring_systems}	Collision, security breach {adversarial_drone_collision_with_train}	Implement full-coverage detection within 200m, <2s latency, 95% confidence.
H2: failure to trigger alerts or responses	Alert not triggered after detection {alert_systems}	Delayed police/train response {inaccurate_response_timing}	Alert must be issued in <1s and acknowledged by control staff

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H3: communication failures with railway control	Communication with {moving_train} traffic controller fails {reliable_railways_traffic_control}	Unsafe {moving_train} movement	Comms must be redundant (N+1), with <200ms failover
H4: Insufficient police authority over drones	{police_response_unit} was not effective due to external situations	Unresolved threat {roaming_adversarial_drone}	(Contextual) all_trains_operations depend on external law enforcement response capabilities

b) Step 2: Building the abstract functional control structure

In this model, we'll represent the hierarchical control structure for the track zone safety controller (TZSC) and its related components, breaking down control loops to identify where authority, feedback, control actions, and interactions occur to support the system's goals.

(1) Define system boundary

The boundary defines the entities included and excluded in the STPA analysis of the TZSC.

- **Included:**
 - **{moving_train}**: the vehicle operating within the track zone.
 - **{clear_train_tracks_zone_open_space}**: the designated ground and airspace surrounding the track area that the TZSC monitors.
 - **{police_force_existing_capability}**: authorised personnel responsible for handling security breaches in the track zone.
 - **{reliable_railways_traffic_control}**: monitors and directs moving_train movements, coordinating with the TZSC for safe moving_train passage.
- **Excluded:**
 - **Passenger reactions** {exclude_passenger_reaction} to delays or disruptions.
 - Operations outside the immediate clear_train_tracks_zone_open_space {exclude_outer_tracks_region_hazards}, such as adjacent urban airspace.
 - Unrelated drone activities outside the scope of railway safety concerns {exclude_other_drone_flying_activities}.

(2) Define the relevant controllers

- **TZSC system controller**: manages and controls access to the clear_train_tracks_zone_open_space and ground zone. It detects unauthorised drones, triggers alerts, and coordinates with police and railway traffic control.

- **reliable_railways_traffic_control:** this system controls all_trains_operations and coordinates with the TZSC to ensure that trains operate within a safe track zone.
- **police_force:** authorised personnel who act upon alerts from the TZSC to neutralise threats.
- **Surveillance system [available_ground_based_security_systems]:** this system monitors and feeds real-time data about drone activity in the track zone to the TZSC.

(3) Model control structure

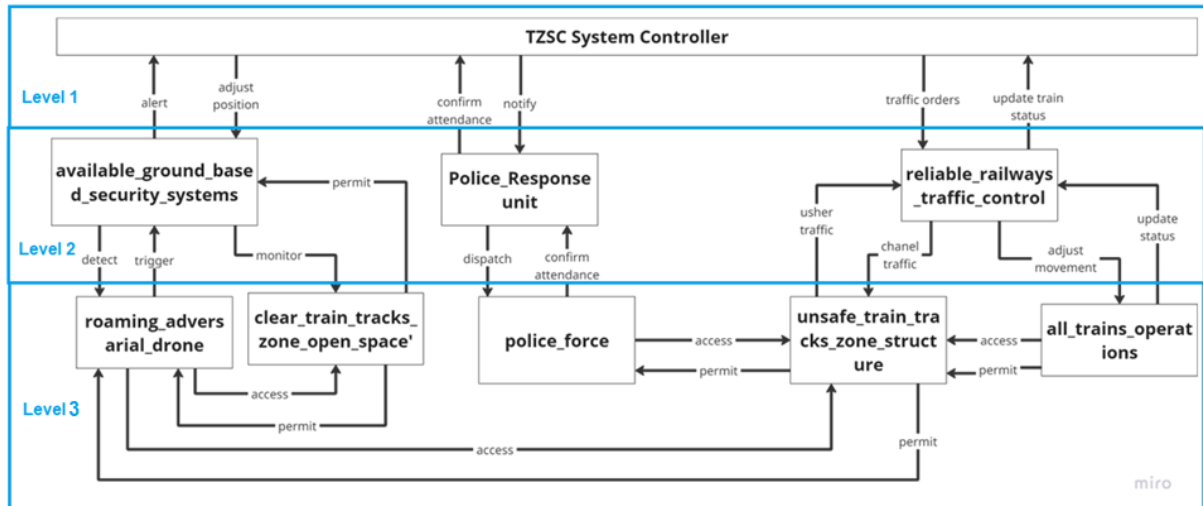


Figure G.4 track zone control structure

(a) Level 1: TZSC system controller (top-level controller)

The {TZSC_system_controller} is the primary authority responsible for overseeing and coordinating actions to maintain safety within the track zone. This controller manages:

- Detection and surveillance of unauthorised drones.
- Communication with the reliable_railways_traffic_control and police_force.
- Coordinate safety responses to ensure trains' safe operation and infrastructure protection.

(b) Level 2: supporting controllers

1. {reliable_railways_traffic_control}
 - Manages moving_train movements within the track zone and works in coordination with the TZSC system controller to maintain operational safety by halting or rerouting trains if a drone threat is detected.
2. {police_response_unit}
 - Responds to alerts from the TZSC to assess and neutralize drone threats in the track zone.
3. {available_ground_based_security_systems}
 - Provides real-time data on drone activity in the track zone to the TZSC system controller, supporting early detection and monitoring.

(c) Level 3: controlled elements

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1. {roaming_adversarial_drone}
 - The unauthorised drone risks the track zone by entering controlled airspace or accessing ground areas.
2. {clear_train_tracks_zone_open_space}
 - The designated airspace is monitored and controlled by the TZSC for intrusion prevention.
3. {police_force_existing_capability}
 - Field personnel who respond to the scene upon alerts from the police_force and take actions to mitigate drone threats.
4. {unsafe_train_tracks_zone_structure}
 - The physical ground area of the track zone, controlled to prevent unauthorized access.
5. {all_trains_operations}
 - The trains operating within the track zone, whose movements must be controlled to avoid collision risks in the presence of a detected drone threat.

(d) Control actions:

Table G.6 STAMP List of control actions

1.	{available_ground_based_security_systems}	detects	{roaming_adversarial_drone}
2.	{available_ground_based_security_systems} alerts {TZSC system controller}		
3.	{available_ground_based_security_systems}	monitors	{clear_train_tracks_zone_open_space}
4.	{roaming_adversarial_drone}	triggers	{available_ground_based_security_systems}
5.	{roaming_adversarial_drone} access {unsafe_train_tracks_zone_structure}		
6.	{roaming_adversarial_drone} accesses {clear_train_tracks_zone_open_space} (above human reach)		
7.	{TZSC system controller} orders {reliable_railways_traffic_control}		
8.	{TZSC system controller} notifies {police_force_existing_capability}		
9.	{TZSC_system_controller}	adjusts	position of {available_ground_based_security_systems}
10.	{police_response_unit} dispatches {police_force_existing_capability}		
11.	{police_force_existing_capability} confirms attendance to {TZSC system controller}		
12.	{police_force_existing_capability}	confirm	attendance to {police_force_existing_capability}

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13.	{police_force_existing_capability}	access	{unsafe_train_tracks_zone_structure}
14.	{all_trains_operations}	access	{unsafe_train_tracks_zone_structure}
15.	{all_trains_operations}	update status to	{reliable_railways_traffic_control}
16.	{reliable_railways_traffic_control}	updates	{TZSC system controller}
17.	{reliable_railways_traffic_control}	adjusts the movement of	{all_trains_operations}
18.	{reliable_railways_traffic_control}	channels	{unsafe_train_tracks_zone_structure}
19.	{unsafe_train_tracks_zone_structure},	usher traffic for the	{reliable_railways_traffic_control}
20.	{unsafe_train_tracks_zone_structure}	permits	{all_trains_operations}
21.	{unsafe_train_tracks_zone_structure}	permits	{police_force_existing_capability}
22.	{unsafe_train_tracks_zone_structure}	permits	{roaming_adversarial_drone}
23.	{clear_train_tracks_zone_open_space}	permits	{roaming_adversarial_drone}
24.	{clear_train_tracks_zone_open_space}	permits	{available_ground_based_security_systems}

G.2.4 Phase 2: Unsafe Control Actions

For each control action (e.g., "Apply brakes"), evaluate:

1. **Not provided** when required.
2. **Provided** when not required.
3. **Too early/too late/too long/too short.**

c) Step 1: Define deviations of control actions

Control action analysis and identification of unsafe actions. We consider scenarios where unsafe actions could lead to system failures or accidents for each control action. Table G.7 captures control action analysis and identification of unsafe actions. The Table evaluates each control action based on scenarios where unsafe actions could lead to failures or accidents. Each row describes a control action, and each column considers potential outcomes.

Table G.7 STAMP control action deviation analysis

Control action	What can go wrong if issued?	What can go wrong if not issued?	Issued too soon/late?	Issued for too long/short?

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1	False positives could lead to unnecessary system responses. {false_positives},	Real drone threats remain undetected, risking security. {false_negatives}	Late detection may lead to an accident. {inaccurate_response_timing}	N/a
2	N/a	Real drone threats remain undetected {false_negatives}	Late alert may lead to an accident. {inaccurate_response_timing}	N/a
3	N/a	Real drone threats remain undetected {false_negatives}	N/a	Short monitoring intervals may miss drone re-entry {adversarial_drone_collision_with_train}, {track_zone_monitoring_systems_data_gaps_or_inconsistencies}
4	N/a	Real drone threats remain undetected, {false_negatives}	Late alert may lead to an accident. {inaccurate_response_timing} {alert_systems}	N/a
5	Drones pose a threat to ground infrastructure and personnel. {unsafe_train_	N/a	N/a	N/a

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	tracks_zone_structure} {safe_secure_police_officer}			
6	Drones pose a threat to ground infrastructure and personnel. {unsafe_train_tracks_zone_structure} {safe_secure_police_officer}	N/a	N/a	N/a
7	N/a	Trains continue despite drone presence, risking collisions or other accidents. {adversarial_drone_collision_with_train}	late orders increase accident risk. {inaccurate_response_timing}	N/a
8	N/a	Delayed notification leaves the drone threat unaddressed, risking escalation. {adversarial_drone_collision_with_train}	Late alert the police may lead to an accident. {inaccurate_response_timing}	N/a
9	Issued inaccurately, Inaccurate Position adjustment can reduce coverage of critical areas. {inaccurate_	Not issued as needed, {track_zone_monitoring_blind_spot}	N/a	N/a

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	monitoring_cove rage}			
10	Unnecessary dispatch strains police resources. {insuffecient_ response_resour ces}	Delayed dispatch leads to a slower response, risking escalation. {inaccurate_ response_time}	N/a	Brief deployment may fail to address threat; prolonged strains resources. {insuffecient_short_d eployment}, {prolonged_ deployment_ strains_resources}
11	False confirmation could lead to inadequate management of the intruding situation. {unmanaged_intr usion_ situation}, {train_delay}.	TZSC is unaware of the response status, risking coordination. {unmanaged_ intrusion_situation}, {train_delay}.	Early confirmation risks miscommunic ation: late confirmation increases uncertainty. {unmanaged_ intrusion_ situation}, {train_delay}.	Short confirmation may not fully verify attendance; long confirmation may delay other actions. {unmanaged_ _intrusion_ _situation}, {train_delay}.
12	False confirmation could disrupt response timing. {unmanaged_intr usion_ situation}, {train_delay}.	Police unit unaware of actual officer presence, risking coordination breakdown. {unmanaged_ intrusion_ situation}, {train_delay}.	Early confirmation may lead to misalignment; late confirmation causes delays in response. {unmanaged_ intrusion_ situation},	Short confirmation may cause insufficient information relay, and long delays. {unmanaged_ intrusion_ situation}, {train_delay}.

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			{train_delay}.	
13	Misinterpretation could lead to traffic interference with {all_trains_operations}, {train_delay}, {safe_secure_police_officer}.	{safe_secure_police_officer}	Early access disrupts {all_trains_operations}; late risks drone escape.{adversarial_drone_collision – with_train}	Brief access may hinder response; long access interferes with moving_train movement. {unmanaged_intrusion_situation}, {train_delay}.
14	Uncontrolled access risks collision with personnel or police in the zone. {adversarial_drone_collision_with_train}, {all_trains_operations}, {train_delay},	N/a	N/a	N/a
15	N/a	Railways Traffic Controller lacks updated status, risking {moving_train} safety and efficiency.{train_schedule_timing}	late updates reduce rtc's situational awareness. {train_schedule_timing},	N/a

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			{reliable_railways_traffic_control}	
16	N/a	TZSC lacks updated status, risking ineffective decision-making. .{train_delay}	late updates reduce situational awareness. {reliable_railways_traffic_control}	N/a
17	N/a	Delayed adjustments risk collision or accident. . {train_delay}, {reliable_railways_traffic_control}	N/a	N/a
18	N/a	N/a	Late control may lead to loss of income and penalties to the {moving_train} operator. {lost_revenue}	N/a
19	N/a	Lack of ushering (unclear or blocked signalling) risks collisions with unauthorised vehicles in the zone.	N/a	N/a

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		.{train_delay}, {adversarial_drone_collision_with_train}		
20	N/a	Trains are unable to access the zones needed for operations. .{train_schedule_timing}, {inaccessible_track_zone} {adversarial_drone_collision_with_train}	N/a	N/a
21	Impact on safety of police officers {safe_secure_police_officer}	Officers are unable to access ground zones needed for response. {train_schedule_timing}, {inaccessible_track_zone} {adversarial_drone_collision_with_train}	N/a	A brief permit may not enable full response or prolonged interference with operations. {unmanaged_intrusion_situation}, {train_delay}.
22	Allowing access would compromise safety and increase collision risk. {train_delay}, {inaccessible_track_zone} {adversarial_drone}	N/a	N/a	N/a

	_collision_with_train}			
23	Allowing access would increase the risk of collisions with trains or police units. {train_delay}, {inaccessible_track_zone} {adversarial_drone_collision_with_train}	N/a	N/a	N/a
24	N/a	Insufficient permits limit airspace monitoring and detection ability. {false_negatives}	N/a	N/a

G.2.5 Phase 3: Causal Analysis

a) **Step 1: Identify causal scenario.**

b) **Step 2: Define safety constraints and requirements**

For each deviation, determine **why** it occurs using:

- **Controller Issues:** Flawed algorithms, incorrect sensor data.
- **Actuator Failures:** Brake malfunction.
- **Sensor Errors:** Faulty GPS signal.
- **Human Situations:** Operator misjudgement.
- **External Disturbances:** Slippery Road.

This section will consider only one action to examine the scenario that led to the unsafe control action:

3. available_ground_based_security_systems monitors clear_train_tracks_zone_open_space

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Below is a Table G.8 that examines the causes and safety constraints:

Table G.8 STAMP Unsafe actions and safety constraints

Unsafe action	Causes [problem articulation]	Mitigation (safety constraints)[solution]
False detections clutter the system with irrelevant data	Sensor malfunctions {monitoring_sensor_malfunction}, environmental interferences (e.g., birds, weather), and poor sensor calibration. {bad_weather}	- implement advanced filtering algorithms to distinguish drones from false positives.
		- regularly calibrate and maintain sensors to minimise errors.
		- machine learning models are used to enhance differentiation between threats and non-threats.
Unauthorised drones remain undetected in the airspace	Insufficient sensor coverage {false_negatives}, delayed response to detection events {inaccurate_response_time} , and inadequate sensitivity of detection systems.	- expand sensor coverage to eliminate blind spots.
		- implement continuous, real-time detection protocols.
		- increase detection sensitivity while minimising false positives through adaptive sensor calibration.
Late monitoring initiation misses early detections.	Communication delays {police_comms_errors}	- automate monitoring activation linked to proximity alerts.
		- establish protocols for immediate monitoring activation upon potential threat identification.
		- ensure real-time communication and minimise human intervention for quicker response.
Short monitoring intervals may miss drone re-entry; long	Resource allocation constraints, insufficient monitoring capabilities, and incorrect interval configuration {insufficient_response_resources}.	- optimize interval settings based on historical drone activity patterns.
		- balance intervals with resources, using event-based monitoring for more targeted detection.

intervals drain resources	- implement predictive analytics to adjust intervals dynamically based on real-time threat analysis.
------------------------------	--

G.2.6 Predictive attention distribution as guided by STAMP

The STAMP/STPA analysis was conducted following the standard methodology in G.5, focusing on control actions (Table G.6), control action deviation analysis (Table G.7), and unsafe actions with safety constraints (Table G.8). The resulting predictive attention distribution reveals a pattern strikingly similar to the HAZOP analysis, exhibiting characteristics consistent with a Normal PAD regime that systematically concentrates attention on high-salience contributing situations while dismissing or omitting low-salience tail categories.

A notable feature of the STAMP analysis was the large number of dismissed situations marked as "not applicable" (n/a) in Table G.9, where certain deviation types did not meaningfully apply to specific control actions. During data curation, these missed failures were systematically reconstructed using a templated substitution scheme to capture what would have been analysed had the deviation been applicable:

- {n/a_control action_n_go wrong if issued}
- {vcontrol action_n_go wrong if not issued}
- {n/a_control action_n_too soon/late}
- {n/a_control action_n_too long\ short}

The following is the data curated from the analysis:

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Table G.9 STAMP predictive attention distribution

Predicted situation	Tally	Attention%	Originality	Typicality Test	Rational for typicality value	Novelty Test
frequently delayed train	17	12.50	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
unsafe train tracks zone structure	10	7.35	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
highly probable adversarial drone collision with an unsafe moving train	10	7.35	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
ineffectively managed drone intrusion	10	7.35	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
reliable railways traffic control	9	6.62	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
police force existing capability	7	5.15	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
false negatives	7	5.15	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
available ground-based security systems	6	4.41	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
roaming adversarial drone	6	4.41	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
TZSC system controller	6	4.41	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
likely inaccurate response time	6	4.41	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
safe secure police officer	5	3.68	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
clear train tracks zone open space	4	2.94	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
all trains operations	4	2.94	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
uncertain train schedule timing	4	2.94	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
inaccessible tracks zone	4	2.94	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
late surveillance alert	3	2.21	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
impacted all trains operations	3	2.21	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
unsafe moving train	2	1.47	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
police response unit	1	0.74	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
false positives	1	0.74	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
late orders	1	0.74	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
prematurely alerted police	1	0.74	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
inaccurate monitoring coverage	1	0.74	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
tracks zone monitoring blind spot	1	0.74	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
Insufficient ineffective police force dispatch response	1	0.74	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
delayed, ineffective police force dispatch response	1	0.74	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
uncertain police response unit	1	0.74	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
Prolonged deployment strains resources	1	0.74	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
lost revenue	1	0.74	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
bad weather	1	0.74	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
insufficient police response resources	1	0.74	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
n/a control action 3 too soon/late	0	negligible	0			
n/a control action 1 too long\ short	0	negligible	0			

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n/a control action 2 go wrong if issued	0	negligible	0		
n/a control action 2 too long\ short	0	negligible	0		
n/a control action 3 go wrong if issued	0	negligible	0		
n/a control action 5 too soon/late	0	negligible	0		
n/a control action 4 too long\ short	0	negligible	0		
n/a control action 4 go wrong if issued	0	negligible	0		
n/a control action 6 too soon/late	0	negligible	0		
n/a control action 5 too long\ short	0	negligible	0		
n/a control action 5 go wrong if not issued	0	negligible	0		
n/a control action 6 too long\ short	0	negligible	0		
n/a control action 6 go wrong if not issued	0	negligible	0		
n/a control action 7 too long\ short	0	negligible	0		
n/a control action 7 go wrong if issued	0	negligible	0		
n/a control action 9 too soon/late	0	negligible	0		
n/a control action 8 too long\ short	0	negligible	0		
n/a control action 8 go wrong if issued	0	negligible	0		
n/a control action 10 too soon/late	0	negligible	0		
n/a control action 9 too long\ short	0	negligible	0		
n/a control action 14 too long\ short	0	negligible	0		
n/a control action 15 too long\ short	0	negligible	0		
n/a control action 16 too long\ short	0	negligible	0		
n/a control action 14 too soon/late	0	negligible	0		
n/a control action 17 too long\ short	0	negligible	0		
n/a control action 18 too long\ short	0	negligible	0		
n/a control action 14 go wrong if not issued	0	negligible	0		
n/a control action 19 too long\ short	0	negligible	0		
n/a control action 15 go wrong if issued	0	negligible	0		
n/a control action 17 too soon/late	0	negligible	0		
n/a control action 20 too long\ short	0	negligible	0		
n/a control action 16 go wrong if issued	0	negligible	0		
n/a control action 22 too long\ short	0	negligible	0		
n/a control action 17 go wrong if issued	0	negligible	0		
n/a control action 19 too soon/late	0	negligible	0		
n/a control action 23 too long\ short	0	negligible	0		
n/a control action 18 go wrong if issued	0	negligible	0		
n/a control action 18 go wrong if not issued	0	negligible	0		
n/a control action 23 go wrong if not issued	0	negligible	0		

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n/a control action 24 go wrong if issued	0	negligible	0			
n/a control action 20 too soon/late	0	negligible	0			
n/a control action 24 too long\ short	0	negligible	0			
n/a control action 19 go wrong if issued	0	negligible	0			
n/a control action 21 too soon/late	0	negligible	0			
n/a control action 20 go wrong if issued	0	negligible	0			
n/a control action 22 too soon/late	0	negligible	0			
n/a control action 23 too soon/late	0	negligible	0			
n/a control action 22 go wrong if not issued	0	negligible	0			
n/a control action 24 too soon/late	0	negligible	0			

Through this reconstruction process, 49 missed failures were identified as inapplicable and therefore assigned zero predicted situations. This represents a substantial epistemic gap: while the STAMP ontology in principle mandates examination of four deviation types per control action, in practice nearly 62% of the theoretically possible deviation scenarios (49 out of 79 total potential deviations) were deemed irrelevant and excluded from analysis. This exclusion pattern mirrors the zero-bin phenomenon observed in the HAZOP analysis (Table G.4), where 8 guidewords received zero attention.

The computed statistical properties of this distribution are (see figure G.2.6):

- Kurtosis = 0.40 (excess kurtosis, slightly positive but close to zero)
- Skewness = 0.06 (near-perfect symmetry)
- Zero bins present: 49 inapplicable control-action deviations

These metrics place the STAMP predictive attention distribution squarely within the Normal PAD regime ($-1 < \text{kurtosis} \leq 0.5$, skewness ≈ 0 , zero bins present). The near-zero kurtosis indicates a mesokurtic distribution with thin tails, neither heavy-tailed (which would signal sustained attention on rare events) nor extremely platykurtic (which would indicate complete uniformity). The near-zero skewness indicates approximate symmetry around the mean, yet this symmetry is achieved through selective inclusion: 49 theoretically possible deviation scenarios were excluded as inapplicable, effectively truncating the tail of the distribution. The presence of 49 zero bins indicates that entire categories of control-action deviations were excluded from the analysis, mirroring the guideword-neglect pattern observed in HAZOP.

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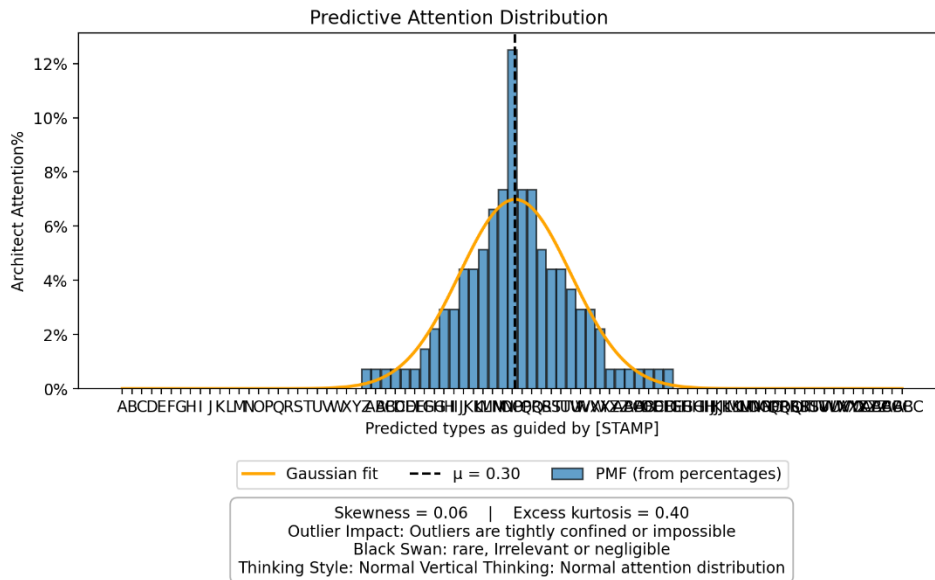


Figure G.2.6 Architect Attention distribution over predicted situations as guided by STAMP process

The STAMP/STPA framework, like HAZOP, exhibits a structural bias toward Normal Vertical Thinking. Despite its ostensibly systematic procedure, enumerate control actions, apply four deviation types per action, derive safety constraints, the method does not enforce exhaustive tail exploration. Instead, it permits (and implicitly encourages) the architect to dismiss deviation types as "not applicable" based on domain intuition and salience judgments.

Compared with HAZOP guide words, STAMP deviations appear more general, encompassing a wider range of interactions. However, they remain a state-change-based ontology where the answers to such changes are either yes or no. They are fewer in number; HAZOP has around 17 types of deviations, while STAMP has only 6.

Similar to our argument for HAZOP, this limitation means that other types of deviations (practically speaking, if the architect adheres to STAMP) are discarded as irrelevant or negligible. Such an attitude towards resolving the predictability of an open and complicated problem space means that key Black Swan scenarios will be overlooked.

G.3 Applying the FRAM method

FRAM enables a more clearly defined solution space for predicting situations. It includes a deeper type system of interactions and deviations. The fo

In this section, we will reuse the problem brief model as we discussed in HAZOP section with the following interactions:

1. moving_train-clear_train_tracks_zone_open_space: the moving_train passes through the clear_train_tracks_zone_open_space.
2. roaming_adversarial_drone – clear_train_tracks_zone_open_space: the roaming_adversarial_drone accesses the clear_train_tracks_zone_open_space.
3. Police_Response_unit – roaming_adversarial_drone: the Police_Response_unit chases the roaming_adversarial_drone.
4. Police_Response_unit – clear_train_tracks_zone_open_space: the Police_Response_unit gains access to the clear_train_tracks_zone_open_space.
5. roaming_adversarial_drone – moving_train : the roaming_adversarial_drone approaches the train.
6. Police_Response_unit – moving_train : the Police_Response_unit denies moving_train access.

The aim is to discover any further complications due to functional resonance that we may not have thought about otherwise in the list of interactions.

G.3.1 Step 1: identify and describe functions

Here's how we can identify and describe the critical functions involved in the given interactions:

- 1) **Function 1: moving_train transit people and goods:** the moving_train operates along its designated track, following set schedules and safety protocols. Its primary function is safely transporting passengers or goods through the clear_train_tracks_zone_open_space.
- 2) **Function 2: clear_train_tracks_zone_open_space facilitating safe passage for trains:** ensures trains can operate smoothly and safely, minimising the risk of obstacles or intrusions that could lead to delays or accidents.
- 3) **Function 3: Police_Response_unit chase roaming_adversarial_drone:** the Police_Response_unit actively pursues the roaming_adversarial_drone, using equipment or tactical manoeuvres to intercept and neutralise the threat posed by the drone.
- 4) **Function 4: roaming_adversarial_drone approach moving_train:** the roaming_adversarial_drone manoeuvres towards the train, possibly to disrupt operations, deliver a payload, or capture sensitive information.

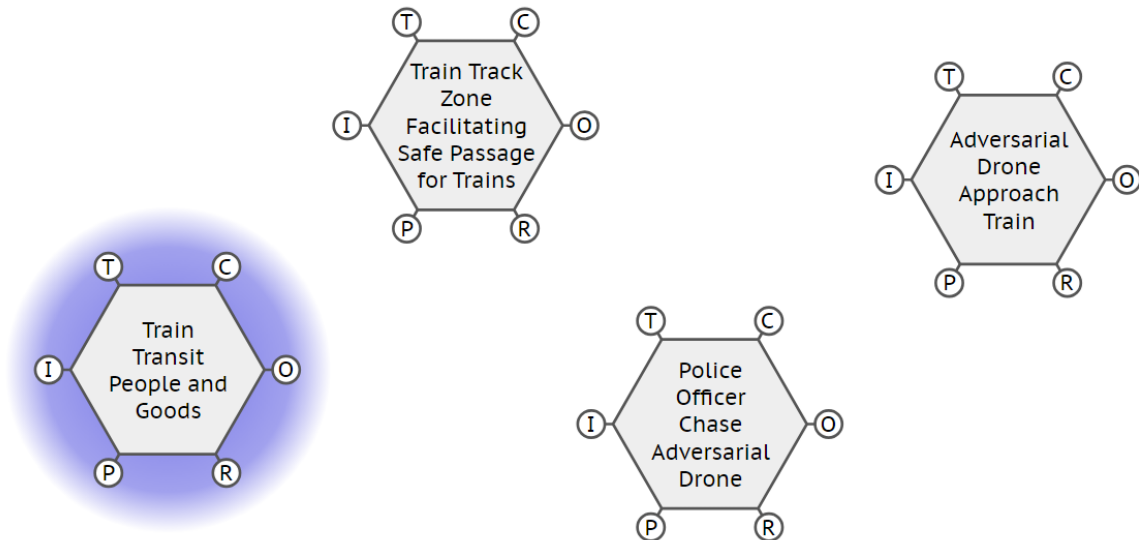


Figure G.5 main functions involved in the problem domain

Then describe each function using six aspects:

Using the six aspects to describe each function in the problem interactions further, here's an in-depth breakdown:

Function 1: moving_train transit people and goods

- **Input (i):** scheduled moving_train passage time and route plan.
- **Output (o):** safe arrival at the destination with passengers and goods.
- **Precondition (p):** clear and secure track for moving_train transit, the moving_train is operational, and safety systems are active.
- **Resource (r):** electric powerlines.
- **Control (c):** moving_train driver, railways traffic control, speed limits, and signal systems.
- **Time (t):** scheduled time of passage through the clear_train_tracks_zone_open_space.

Function 2: clear_train_tracks_zone_open_space facilitating safe passage for trains

- **Input (i):** presence of a train, scheduled moving_train passage.
- **Output (o):** clear and secure track for moving_train transit.
- **Precondition (p):** track inspection is complete, and no unauthorised objects or people are within the zone.
- **Resource (r):** surveillance systems, fencing, track zone infrastructure.
- **Control (c):** railway safety regulations.
- **Time (t):** n/a.

Function 3: Police_Response_unit chase roaming_adversarial_drone

- **Input (i):** alert of a roaming_adversarial_drone in the clear_train_tracks_zone_open_space.
- **Output (o):** neutralisation or removal of the drone threat.

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- **Precondition (p):** the drone's location is detected, and police access to the clear_train_tracks_zone_open_space is granted.
- **Resource (r):** Police_Response_unit, communication systems.
- **Control (c):** law enforcement protocols, safety procedures for high-risk zones.
- **Time (t):** 12 minutes to respond.

Function 4: roaming_adversarial_drone approach train

- **Input (i):** moving_train schedule, route to target track zone plan.
- **Output (o):** collision with train.
- **Precondition (p):** clear clear_train_tracks_zone_open_space, undetected or unimpeded approach.
- **Resource (r):** GPS, well-trained roaming_adversarial_drone perception, functional wireless communication network, full battery life.
- **Control (c):** roaming_adversarial_drone pilot.
- **Time (t):** 30 minutes flight.

a) Defining background functions:

One helpful aspect of FRAM is the definition of background and foreground functions. Usually, the background functions are resources or conditional inputs (givens) or expected outcomes where the assumption is not to define p, r, c, or t. To do so, let's reiterate the problem brief:

An anonymous drone intruded into a bounded clear_train_tracks_zone_open_space while the moving_train passed by. police_force were called to the scene and tried to capture the drone but were unsuccessful.

We could identify the following background functions:

- **Open airspace:** this provides input access to roaming_adversarial_drones and to passing trains and police_force.
- **Passenger safe arrival to destination** is the output of police, clear_train_tracks_zone_open_space, and train.
- **Congested road traffic** impacts the arrival time for police to attend the scene.
- **Environmental conditions:** impact roaming_adversarial_drone success.

We simulated the model after defining background functions and we got the following results:

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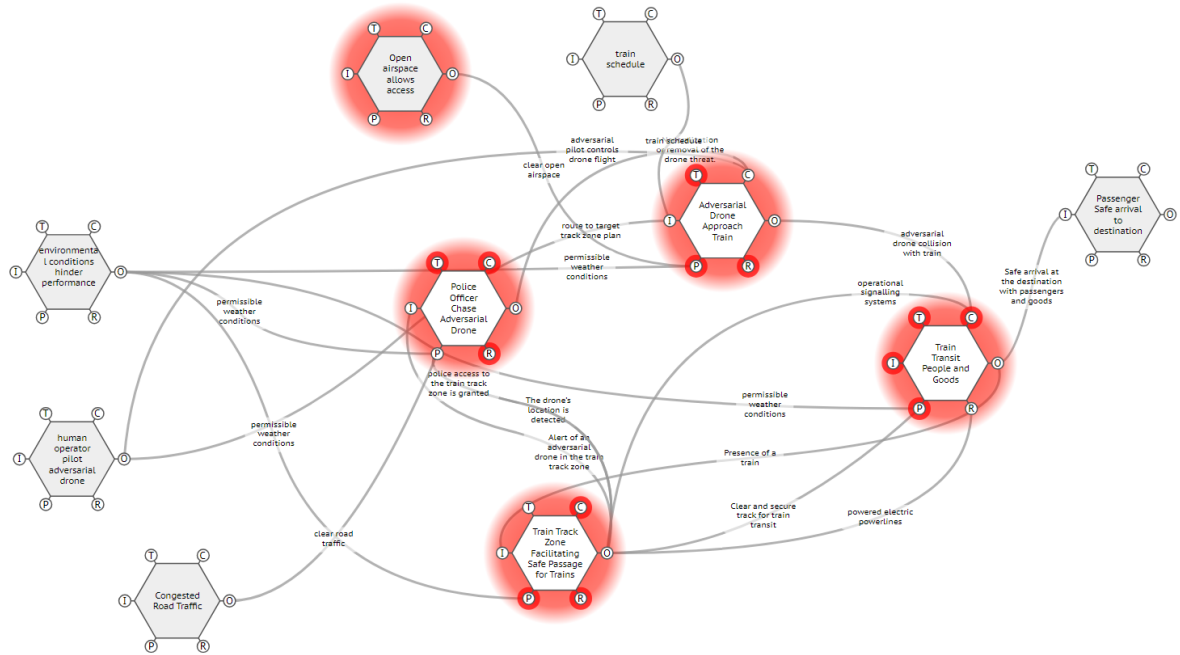


Figure G.6 background functions definition

The initial model of the problem has produced the following errors:

Warning: <roaming_adversarial_drone approach train> has an orphan

Warning: <train transit people and goods> has an orphan

Warning: <Police_Response_unit chase roaming_adversarial_drone> has an orphan

Warning: <clear_train_tracks_zone_open_space facilitating safe passage for trains> has an orphan

In the FRAM model interpreter context, the warning "has an orphan" typically means that the function listed lacks a connection to another function or aspect within the FRAM model. In a FRAM model, each function should have defined inputs, outputs, or dependencies that link it to other functions. When a function is identified as an "orphan," it indicates that:

1. **No downstream dependencies:** the function produces an output, but no downstream functions are using this output.
2. **No upstream inputs:** the function requires inputs, controls, or resources that aren't connected to any other function, suggesting that it might not receive the necessary context or triggering conditions.

Possible implications of orphan functions:

- **Model completeness issue:** the model may be incomplete if the orphan function lacks sufficient connections, as it leaves functional dependencies unaddressed.
- **Impact on functional resonance analysis:** orphan functions can affect the accuracy of the FRAM model's analysis because unconnected functions do not interact with the rest of the system, potentially overlooking critical variability interactions.

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To complete the model, we discovered that all aspects (CRIPTO) of foreground functions must be associated with an output of some influencing function. This helped us identify more interactions we didn't consider initially. By making the adjustments we managed to discharge all errors:

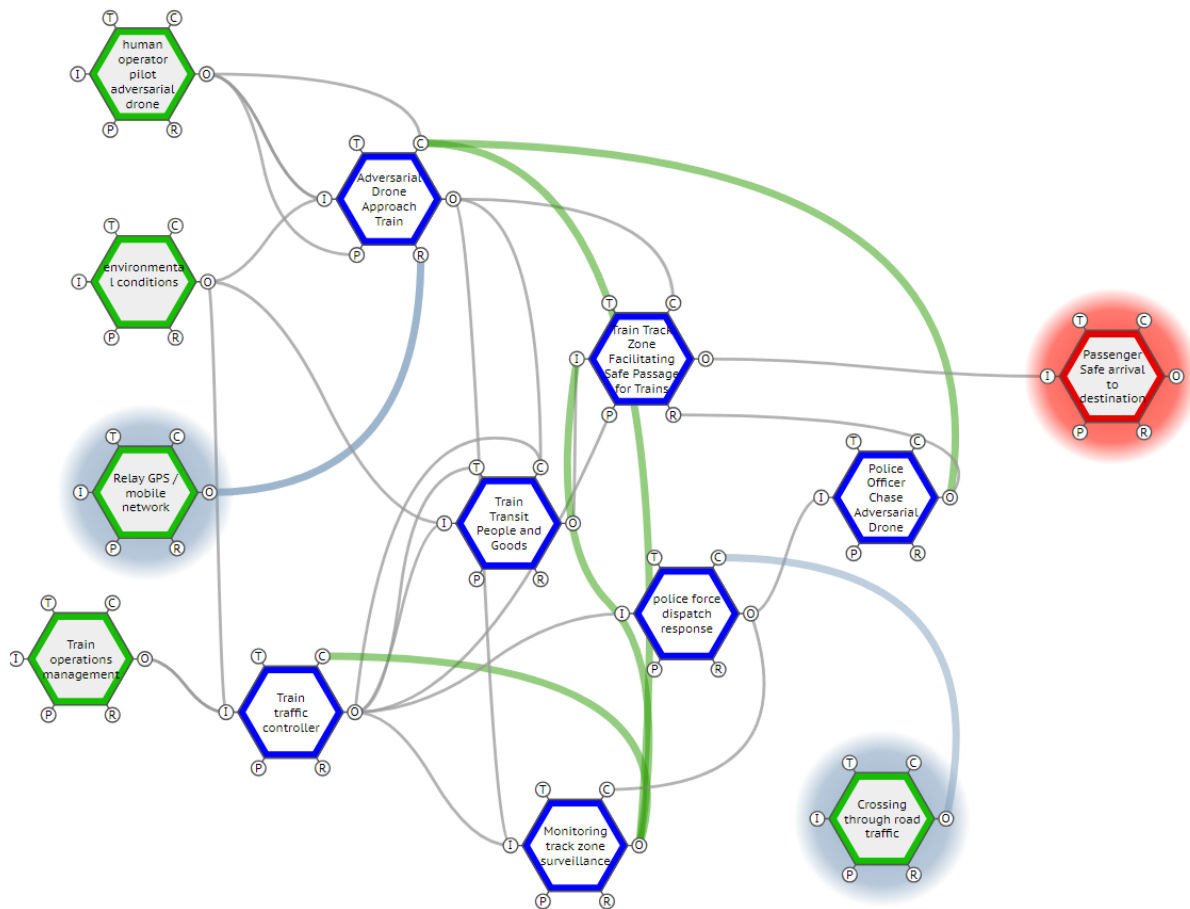


Figure G.7 full problem articulation

The following is the output interpretation of the model which also constitute a FRAM based problem articulation:

FRAM model interpreter - fmi basic ----- (c) erik hollnagel, 2020

Fmi session log: 10/27/2024 11:03:50 pm

Entry function <environmental conditions>

Exit function <passenger safe arrival to destination >

Entry function <human operator pilot roaming_adversarial_drone>

Entry function <all_trains_operations>

Interpretation profile

Function <all_trains_operations>:

output: 4

Function <human operator pilot roaming_adversarial_drone>:

output: 4

Function <environmental conditions>:

output: 4

Function <roaming_adversarial_drone approach train>:

input: all precondition: all resource: all control: none time: all

Function <train transit people and goods>:

input: all precondition: all resource: all control: all time: all

Function <Police_Response_unit chase roaming_adversarial_drone>:

input: all precondition: all resource: all control: all time: all

Function <clear_train_tracks_zone_open_space facilitating safe passage for trains>:

input: any precondition: all resource: all control: all time: all

Function <monitoring track zone surveillance>:

input: all precondition: all resource: all control: any time: all

Function <police_force dispatch response>:

input: all precondition: all resource: all control: all time: all

Function <train traffic controller>:

input: all precondition: all resource: all control: none time: all

Function <passenger safe arrival to destination >:

input: all

Summary of fmi log

Begin initialisation

--- model initialisation completed.

Begin cycle 0

Background function <relay GPS / mobile network> has been activated

Entry function <environmental conditions> has been activated

Background function <crossing through road traffic> has been activated

Entry function <human operator pilot roaming_adversarial_drone> has been activated

Entry function <all_trains_operations> has been activated

Begin cycle 1

Function <roaming_adversarial_drone approach train> has been activated

Function <train traffic controller> has been activated

Begin cycle 2

Function <train transit people and goods> has been activated

Function <police_force dispatch response> has been activated

Begin cycle 3

Function <Police_Response_unit chase roaming_adversarial_drone> has been activated

Function <monitoring track zone surveillance> has been activated

Begin cycle 4

Entry function <environmental conditions> has been activated

Function <clear_train_tracks_zone_open_space facilitating safe passage for trains> has been activated

Entry function <human operator pilot roaming_adversarial_drone> has been activated

Entry function <all_trains_operations> has been activated

Begin cycle 5

Function <roaming_adversarial_drone approach train> has been activated

Exit function <passenger safe arrival to destination > has been activated

Function <train traffic controller> has been activated

Below is the interpretation of the problem articulation output:

1. Core functional interactions and dependencies:

- **all_trains_operations:** this function orchestrates the moving_train schedule and manages communications with moving_train operators, fundamentally influencing the entire moving_train traffic system. It outputs critical scheduling and coordination data to the moving_train traffic controller, setting the operational context.
- **Train traffic controller:** As the primary system overseer, this function guides the train's safe transit through scheduling, monitoring, and controlling passage zones. Dependencies on inputs like surveillance alerts, road traffic data, and fencing ensure secure passage, but make it vulnerable if any input fails or is delayed.
- **Monitoring track zone surveillance:** This function actively tracks the train's transit zone for unauthorised access, triggering alerts that guide responses from both the moving_train traffic controller and police dispatch. Surveillance effectiveness is critical for detecting and mitigating intrusions, especially with a roaming_adversarial_drone threat.

2. Introduction of the roaming_adversarial_drone:

- The function of the **roaming_adversarial_drone approach moving_train** introduces disruptive dynamics, relying on inputs like GPS signals, weather conditions, and control from a human operator (adversarial pilot) with knowledge of moving_train tracks. The drone's capabilities for perception and targeting set the stage for adversarial interference, potentially bypassing or stressing surveillance and response protocols.

3. Police and response coordination:

- **police_force_dispatch response** is activated upon monitoring track zone surveillance alerts, initiating a chase or defensive action against the drone. This function works alongside **Police_Response_unit chase roaming_adversarial_drone**, who captures the drone and secures the track zone. This layer adds resilience by enabling law enforcement to mitigate drone threats physically.

- **Relay GPS / mobile network and crossing through road traffic:** both background functions provide essential environmental data to various primary functions. Any GPS failure or unexpected traffic changes could disrupt not only Police_response_unit times but also the drone's navigational controls, leading to unpredictability's in the operation.

4. Interpretation of the cycles:

This breakdown of cycles demonstrates the activation and interaction of key functions within the moving_train transit system, particularly in the presence of an roaming_adversarial_drone threat. Each cycle reveals how different functions respond and interconnect, contributing to the system's resilience, or points of potential vulnerability.

Cycle 0: initial setup and background conditions

- **Activated functions:** relay GPS / mobile network, environmental conditions, crossing through road traffic, human operator pilot roaming_adversarial_drone, and all_trains_operations.
- **Interpretation:** this initial cycle establishes essential background and entry functions, setting up the operational environment. Relay GPS / mobile network connectivity, and crossing through road traffic, indicate infrastructure support, while environmental conditions prepare the system to account for weather and external physical conditions. Human-operator pilot roaming_adversarial_drone and all_trains_operations entry functions introduce both the drone operator and the moving_train schedule, marking the beginning of potential adversarial influence in the setup.

Cycle 1: drone engagement and moving_train control activation

- **Activated functions:** roaming_adversarial_drone approach, moving_train and moving_train traffic controller.
- **Interpretation:** The roaming_adversarial_drone approach moving_train function is activated, showing the roaming_adversarial_drone entering the system and initiating an approach toward the clear_train_tracks_zone_open_space. This triggers the moving_train traffic controller, responsible for safely guiding moving_train traffic. The system now must contend with managing regular moving_train passage while accounting for the potential risk posed by the drone.

Cycle 2: transit and response readiness

- **Activated functions:** moving_train transit people and goods and police_force dispatch response.
- **Interpretation:** this cycle shows active moving_train movement with moving_train transit people and goods, indicating a scheduled or in-progress transit. police_force dispatch response is activated in response to the drone threat, setting up law enforcement to counter the adversarial presence. This cycle highlights how the system escalates to

include law enforcement as soon as the drone activity is confirmed, showcasing a direct reaction to the security risk.

Cycle 3: drone pursuit and surveillance

- **Activated functions:** Police_Response_unit chases roaming_adversarial_drone and monitoring track zone surveillance.
- **Interpretation:** Here, a Police_Response_unit chasing an roaming_adversarial_drone reflects active measures to pursue or intercept the drone, indicating an attempt to mitigate the drone threat. Monitoring track zone surveillance ensures continuous oversight of the area, particularly important as law enforcement interacts with the drone. Surveillance acts as a control mechanism, tracking both the train's safe transit and the evolving drone situation.

Cycle 4: reassessment and re-initialization of entry functions

- **Activated functions:** environmental conditions, clear_train_tracks_zone_open_space facilitating safe passage for trains, human operator pilot roaming_adversarial_drone, and all_trains_operations.
- **Interpretation:** this cycle involves a refresh of environmental conditions, human operator pilot roaming_adversarial_drone, and all_trains_operations entry functions, possibly due to shifts in the situation or environmental situations. clear_train_tracks_zone_open_space, facilitating safe passage for trains, is also activated, maintaining safe transit by accounting for both the drone threat and real-time environmental conditions. Reinitialising entry functions indicates an adaptive response within the system, aimed at maintaining a stable operational environment.

Cycle 5: final threat management and system continuation

- **Activated functions:** roaming_adversarial_drone approach train, passenger safe arrival to destination, and moving_train traffic controller.
- **Interpretation:** the roaming_adversarial_drone approach moving_train function is triggered again, suggesting an ongoing or recurring drone threat. Despite this, the passenger-safe-arrival-to-destination function is activated, indicating that the moving_train has successfully passed through the affected zone safely. The moving_train traffic controller remains active, ensuring continued oversight. This final cycle indicates that despite adversarial interference, the system achieves its goal of safe passenger arrival.

G.3.2 Step 2: Characterise variability in outputs

The following is the output list of coupled functions:

1. <train traffic controller>[scheduled moving_train passage time] into {time}<train transit people and goods>

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2. <train traffic controller>[guide moving_train driver] into {control}<train transit people and goods>
3. <train traffic controller>[adequate fencing] into {precondition}<clear_train_tracks_zone_open_space facilitating safe passage for trains>
4. <train traffic controller>[monitor track zone surveillance] into {input}<monitoring track zone surveillance>
5. <train traffic controller>[dispatch train] into {input}<train transit people and goods>
6. <monitoring track zone surveillance>[alert unauthorised access] into {control}<train traffic controller>
7. <human operator pilot roaming_adversarial_drone>[pilot roaming_adversarial_drone] into {control}<roaming_adversarial_drone approach train>
8. <human operator pilot roaming_adversarial_drone>[capable adversarial perception model] into {precondition}<roaming_adversarial_drone approach train>
9. <human operator pilot roaming_adversarial_drone>[clear_train_tracks_zone_open_space location] into {input}<roaming_adversarial_drone approach train>
10. <human operator pilot roaming_adversarial_drone>[adversarial mission target] into {input}<roaming_adversarial_drone approach train>
11. <environmental conditions>[permissible weather conditions] into {input}<roaming_adversarial_drone approach train>
12. <environmental conditions>[permissible weather conditions] into {input}<train transit people and goods>
13. <environmental conditions>[permissible weather conditions] into {input}<train traffic controller>
14. <relay GPS / mobile network>[GPS signal] into {resource}<roaming_adversarial_drone approach train>
15. <relay GPS / mobile network>[provide mobile network] into {resource}<roaming_adversarial_drone approach train>
16. <roaming_adversarial_drone approach train>[roam clear_train_tracks_zone_open_space] into {control}<clear_train_tracks_zone_open_space facilitating safe passage for trains>
17. <train transit people and goods>[passing through track zone] into {input}<clear_train_tracks_zone_open_space facilitating safe passage for trains>
18. <monitoring track zone surveillance>[passing through track zone] into {input}<clear_train_tracks_zone_open_space facilitating safe passage for trains>
19. <train traffic controller>[alert police_force] into {input}<police_force dispatch response>

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20. <police_force dispatch response>[dispatch police_force] into {input}<Police_Response_unit chase roaming_adversarial_drone>
21. <Police_Response_unit chase roaming_adversarial_drone>[capture roaming_adversarial_drone] into {control}<roaming_adversarial_drone approach train>
22. <crossing through road traffic>[permissible road traffic] into {control}<police_force dispatch response>
23. <clear_train_tracks_zone_open_space facilitating safe passage for trains>[safe transport of people] into {input}<passenger safe arrival to destination >
24. <Police_Response_unit chase roaming_adversarial_drone>[secure clear_train_tracks_zone_open_space] into {resource}<clear_train_tracks_zone_open_space facilitating safe passage for trains>
25. <police_force dispatch response>[police monitor track zone surveillance] into {control}<monitoring track zone surveillance>
26. <all_trains_operations>[train schedule management] into {input}<train traffic controller>
27. <all_trains_operations>[communication with moving_train operators] into {input}<train traffic controller>
28. <roaming_adversarial_drone approach train>[miss moving_train collision] into {control}<train transit people and goods>
29. <roaming_adversarial_drone approach train>[trigger surveillance alert] into {input}<monitoring track zone surveillance>
30. <monitoring track zone surveillance>[detect roaming_adversarial_drone] into {control}<roaming_adversarial_drone approach train>

Then we need to define deviations for each coupled function:

Table G.10 FRAM deviation of the problem domain

Coupled functions scenario	Possible variability scenarios			
	Temporal	Object	Direction	Distance
<upstream> output [output] into aspect of <downstream>	Consider internal or external variability to cause the impact of time variability.	Consider internal or external variability to cause the impact of object variability.	Consider internal or external variability to cause the impact of direction variability.	Consider internal or external variability to cause the impact of distance variability.
<train traffic controller> output [scheduled moving_train passage time] into time situation of <train transit people and goods>	Too late: the schedule has been delayed, affecting coordination with other scheduled_moving_train passages. {delayed_scheduled_moving_train passages}	Wrong moving_train: incorrect moving_train information is scheduled, impacting other transit schedules.{wrong_scheduled_moving_train passages}	Wrong route: the scheduled time reflects an incorrect route for the train, misaligning with traffic flow.{wrong_route_scheduled_moving_train_passages}	N/a
<train traffic controller> output [guide moving_train driver] into control situation of <train transit	Too early: guidance is given too early, confusing the driver about timing. {early_guidance_to_moving_train_driver} Too late: delayed guidance disrupts coordinated all_trains_operations.{late_guidan	Wrong moving_train driver: the guidance is mistakenly sent to a different moving_train driver, impacting both operations.{wrong_guidance_recipient_moving_train_driver}	Wrong direction: instructions are provided for an incorrect track, potentially leading the moving_train off-course.{wrong_direct	N/a

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people and goods>	ce_affecting_all_trains_operations }		ion_guidance_for_moving_train}	
<train traffic controller> output [adequate fencing] into precondition situation of <clear_train_tracks_zone_open_space facilitating safe passage for trains>	n/a	Inadequate fencing: incorrect type or insufficient fencing materials reduce safety in the transit zone.{inadequate_track_zone_fencing}	Fencing in wrong location: fencing is reinforced in a non-critical area, leaving track zones vulnerable.{misplaced_track_zone_fencing}	Too short: adequate fencing only covers part of the zone, failing to protect the entire transit area.{partial_coverage_track_zone_fencing}
<train traffic controller> output [monitor track zone surveillance] into input situation of <monitoring track zone surveillance>	Too late: surveillance begins after the moving_train is already in the zone, potentially missing threats.{late_track_zone_surveillance}	Incorrect zone: surveillance is directed to a different zone, leaving the target area unmonitored.{wrong_track_zone_surveillance}	Wrong orientation: surveillance cameras are oriented in the wrong direction, missing key areas.{misoriented_track_zone_surveillance_cameras}	Too short: surveillance time is brief, failing to monitor the train's entire passage.{short_duration_track_zone_surveillance}
<train traffic controller> output [dispatch train] into input situation of <train transit people and goods>	Too early: moving_train dispatch occurs before passengers have boarded, causing inconvenience and delays.{early_moving_train_dispatch} Too late: moving_train is dispatched late, delaying all subsequent schedules.{late_moving_train_dispatch}	Wrong moving_train dispatched: an incorrect moving_train is sent, disrupting the moving_train system schedule.{wrong_moving_train_dispatch}	Wrong direction: the moving_train is dispatched onto an incorrect track, leading to misdirection.{wrong_track_moving_train_dispatch}	Too long: dispatch includes extended segments beyond its intended destination, causing inefficiency.{overextended_moving_train_dispatch}
<monitoring track zone surveillance> output [alert unauthorised access] into control situation of <train traffic controller>	Too early: alert is triggered before unauthorized access occurs, leading to unnecessary resource allocation.{early_unauthorised_access_alert} Too late: alert comes after unauthorized access has already affected all_trains_operations.{late_unauthorised_access_alert_affecting_all_trains_operations}	False alert: the system misidentifies authorised personnel as unauthorised, causing operational delays.{false_unauthorised_access_alert}	Incorrect alert zone: alert is sent for an unrelated track zone, missing the target area of unauthorized access.{wrong_track_zone_unauthorised_access_alert}	Alert range too short: alert fails to cover the entire track zone, missing parts of unauthorized access areas.{short_range_unauthorised_access_alert}
<human operator pilot roaming_adversarial_drone> output [pilot roaming_adversarial_drone] into control situation of <roaming_adversarial_drone approach train>	n/a	n/a	n/a	n/a
<human operator pilot roaming_adversarial_drone> output [capable adversarial perception model] into precondition situation of <roaming_adversarial_drone approach train>	n/a	n/a	n/a	n/a
<human operator pilot	n/a	n/a	n/a	n/a

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roaming_adversarial_drone> output [clear_train_tracks_zone_open_space location] into input situation of <roaming_adversarial_drone approach train>				
<human operator pilot roaming_adversarial_drone> output [adversarial mission target] into input situation of <roaming_adversarial_drone approach train>	n/a	n/a	n/a	n/a
<environmental conditions> output [permissible weather conditions] into input situation of <roaming_adversarial_drone approach train>	Too early: {environmental conditions permit adversarial drone operation earlier than expected}, allowing the drone to advance into the moving_train zone before adequate countermeasures are in place.	N/a	N/a	N/a
<environmental conditions> output [permissible weather conditions] into input situation of <train transit people and goods>	Too late: {updated weather conditions are reported late} after the moving_train has already started transit, increasing the chance of adverse effects en route.	Incorrect forecast: {incorrect weather forecast} forecasts poor conditions, leading to unnecessary delays, while the real threat (roaming_adversarial_drone) remains undetected.	Wrong focus area: {weather reports focus on irrelevant areas} adjacent to the train's path, missing critical conditions in the train's route.	N/a
<environmental conditions> output [permissible weather conditions] into input situation of <train traffic controller>	N/a	{Wrong weather conditions reported to train traffic controller}: misreported conditions prompt unnecessary delays or actions, leaving potential threats unmonitored.	{Wrong weather zone analysed by traffic controller}: the controller analysis weather information for the wrong zone, misaligning control actions for the actual conditions faced by the train.	N/a
<relay GPS / mobile network> output [GPS signal] into resource situation of <roaming_adversarial_drone approach train>	Too late: {delayed GPS signal} disrupts the drone's timing, potentially causing it to miss or inaccurately target the train, risking broader security concerns.	Incorrect signal source: {signal data from an incorrect source or spoofed signal} leads the drone to the wrong location but still within reach of the moving_train route.	N/a	N/a
<relay GPS / mobile network> output [provide mobile	N/a	N/a	N/a	N/a

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network] into resource situation of <roaming_adversarial_drone approach train>				
<roaming_adversarial_drone approach train> output [roam clear_train_tracks_zone_open_space] into control situation of <clear_train_tracks_zone_open_space facilitating safe passage for trains>	Too early: {adversarial drone begins roaming before a moving_train is present}, enabling it to scout for weak points and increasing collision potential.	N/a	N/a	Too wide: {adversarial drone covers wider area} roams an extended area beyond the immediate track zone, escalating risks to adjacent all_trains_operations.
<train transit people and goods> output [passing through track zone] into input situation of <clear_train_tracks_zone_open_space facilitating safe passage for trains>	Too late: {moving_train arrival delays complicate coordination} and increase the chance of collision with an roaming_adversarial_drone lingering in the zone.	N/a	N/a	N/a
<train traffic controller> output [alert police_force] into input situation of <police_force dispatch response>	Too early: {police are alerted prematurely} before any threat fully materialises, risking resource misallocation and response fatigue. Too late: the {late alert of adversarial drone presence} comes after an incident has escalated, delaying Police_response_unit and increasing risk to moving_train and track safety.	Incorrect alert target: police are dispatched to the wrong area, leaving the true threat zone unprotected. {wrong police dispatch}	N/a	N/a
<police_force dispatch response> output [dispatch police_force] into input situation of <Police_Response_unit chase roaming_adversarial_drone>	N/a	Wrong unit: a {less equipped police unit is dispatched}, limiting the response capacity against the roaming_adversarial_drone.	Wrong direction: {police officers' approach wrong zone} route, reducing their efficiency in intercepting the drone's path.	Too far: {too far dispatch location of officers} are dispatched from too far station from the track zone area, wasting resources and response time.
<Police_Response_unit chase roaming_adversarial_drone> output [capture roaming_adversarial_drone] into control situation of <roaming_adversarial_drone approach train>	Too late: capture attempt happens after the drone completes its mission, failing to prevent adversarial actions. {late chase of adversarial drone}	N/a	N/a	Too far: officers pursue the drone too far from the track zone, losing coverage of other vulnerable areas. {excessive police chasing distance}
<crossing through road traffic> output	Too late: traffic control updates are delayed, causing police to be stuck in traffic when urgent	N/a	N/a.	N/a

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[permissible road traffic] into control situation of <police_force dispatch response>	response is needed. {road traffic related delay of police dispatch}			
<clear_train_tracks_zone_open_space facilitating safe passage for trains> output [safe transport of people] into input situation of <passenger safe arrival to destination >	N/a	N/a	N/a	N/a
<Police_Response_unit chase roaming_adversarial_drone> output [secure clear_train_tracks_zone_open_space] into resource situation of <clear_train_tracks_zone_open_space facilitating safe passage for trains>	N/a	N/a	N/a	N/a
<police_force dispatch response> output [police monitor track zone surveillance] into control situation of <monitoring track zone surveillance>	Too late: monitoring starts after an incident has already escalated, reducing the effectiveness of the response. {late monitoring after escalation}	Incorrect surveillance focus: monitoring is directed toward a less vulnerable zone, leaving critical areas exposed. {poor allocation of security monitoring resources}	Wrong orientation: surveillance is focused in the wrong direction, allowing threats to go undetected or poorly addressed. {poor allocation of security monitoring resources}	Too short: surveillance monitoring covers only a portion of the track zone, potentially leaving other areas unmonitored. {poor allocation of security monitoring resources}
<all_trains_operations> output [train schedule management] into input situation of <train traffic controller>	Too late: schedule updates are delayed, leading to unsynchronised moving_train dispatches, which may cause delays or missed connections. {unsynchronised moving_train dispatches}	Incorrect schedule: an outdated or incorrect schedule is provided, causing cascading delays and mismanagement across the transit system. {unsynchronised moving_train dispatches}	Wrong timing focus: scheduling fails to account for rush hours or peak times, leading to congestion and operational inefficiencies. {unsynchronised moving_train dispatches}	N/a
<all_trains_operations> output [communication with moving_train operators] into input situation of <train traffic controller>	Too late: communication delays cause misalignment in coordination, with potential safety risks if operators aren't fully informed. {mismanaged communication leading to unsynchronised moving_train dispatches}	Incorrect communication content: misleading or incorrect information is conveyed to the moving_train operators, risking operational or safety-related errors. {mismanaged communication leading to unsynchronised moving_train dispatches}	Wrong communication direction: information is relayed to the wrong team or department, causing unnecessary action or inaction in key areas. {mismanaged communication leading to unsynchronised moving_train dispatches}	Too limited: communication fails to reach all necessary operators, resulting in fragmented understanding or inconsistent operational execution. {mismanaged communication leading to unsynchronised moving_train dispatches}
<roaming_adversarial_drone approach	N/a	N/a	N/a	N/a

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train> output [miss moving_train collision] into control situation of <train transit people and goods>				
<roaming_adversarial_drone approach train> output [trigger surveillance alert] into input situation of <monitoring track zone surveillance>	Too late: the alert is triggered after the drone has entered the track zone, reducing response time and increasing safety risks. {late surveillance alert}	Wrong target alert: the alert is mistakenly triggered for a different environmental object, causing distraction and possible resource misallocation. {false positive surveillance alert}	Wrong orientation: the alert directs security focus to the wrong area, such as an unrelated section of the track zone, delaying interception. {misdirecting surveillance alert}	Too short: the alert coverage is limited, leaving critical areas of the track zone unmonitored, increasing the risk of undetected unauthorised access. {too short surveillance alert}
<monitoring track zone surveillance> output [detect roaming_adversarial_drone] into control situation of <roaming_adversarial_drone approach train>	N/a	N/a	N/a	N/a

G.3.3 Step 3: Identify functional resonance

Identify functional resonance across non-direct coupled functions

Table G.11 definition of functional resonance

Resonating functions variabilities	Emergent resonance scenario
Negative variability of <train traffic controller>[scheduled moving_train passage time] into {time}<train transit people and goods> & <train traffic controller>[dispatch train] into {input}<train transit people and goods> Resonating with: positive variability of <roaming_adversarial_drone approach train>[miss moving_train collision] into {control}<train transit people and goods> Resulting in:	Given that: Too late: the schedule is delayed, impacting the coordination with other scheduled moving_train passages. {delayed moving_train schedule} Wrong moving_train: incorrect moving_train information is scheduled, impacting other transit schedules. {Incorrect moving train information broadcasted} Wrong route: the scheduled time reflects an incorrect route for the train, misaligning with traffic flow. {Incorrect moving train route information broadcasted} Then:

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Increased the likelihood of an roaming_adversarial_drone missing a moving_train collision.	Leading to a change in moving_train passage through the clear_train_tracks_zone_open_space, thus reducing the impact on the adversarial pilot's expected timing.
Negative variability of <environmental conditions> output [permissible weather conditions] into input situation of <train transit people and goods> Resonating with: Negative variability of <roaming_adversarial_drone approach train>[miss moving_train collision] into {control}<train transit people and goods> Resulting in: Increased the likelihood of roaming_adversarial_drone moving_train collision.	Given that: Incorrect forecast: suppose the forecast anticipated adverse conditions; however, good conditions emerged earlier than expected. Then: This shift may influence the likelihood of an adversarial attack, as the sudden change could suggest an increased probability of such an event occurring.
Negative variability of <environmental conditions> output [permissible weather conditions] into input situation of <train transit people and goods> Resonating with: Negative variability of <crossing through road traffic> output [permissible road traffic] into control situation of <police_force dispatch response> Resonating with: Negative variability of <roaming_adversarial_drone approach train>[miss moving_train collision] into {control}<train transit people and goods> Resulting in:	Given that: Incorrect forecast: suppose the forecast anticipated good conditions, And police ignore road traffic condition since there is no bad weather to cause bad traffic (assuming police tacitly associate good weather with easy traffic), forecast, But the weather changes dramatically, leading to unexpected traffic congestion, Too late: traffic control updates are delayed, causing police to be dispatched when urgent response is needed during bad traffic. Then:

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Increased the likelihood of roaming_adversarial_drone moving_train collision.	This will lead to an increase in the likelihood of roaming_adversarial_drone moving_train collisions.
Negative variability of <train traffic controller> output [monitor track zone surveillance] into input situation of <monitoring track zone surveillance> Resonating with: Negative variability of <train traffic controller> output [adequate fencing] into precondition situation of <clear_train_tracks_zone_open_space facilitating safe passage for trains> Resonating with: Negative variability of <roaming_adversarial_drone approach train> output [trigger surveillance alert] into input situation of <monitoring track zone surveillance>	Given that: Wrong orientation: surveillance cameras are oriented in the wrong direction, missing key areas. Inadequate fencing: incorrect type or insufficient fencing materials reduce safety in the transit zone. Then: The fence's inadequacy could impact the surveillance system, which is untrained to consider the fence shape. An roaming_adversarial_drone approach may lead to a false positive or negatives.

G.3.4 Predictive Attention Distribution as guided by FRAM

We applied attention distribution analysis only to the output content of Section G.3.2. We converted the interaction list entities (< >, []) to situation format { } and included predicted situations in the deviation scenarios in Table G.10. Then applied the attention analysis approach in Section G.5. For values that are labelled as “n/a”, we identified them as “n/a_FRAM_[type]_deviation_n” , for example:

- n/a_FRAM_direction_deviation_23
- n/a_FRAM_object_deviation_13

The analysis helped us to predict 92 types of situations, but dismissed 61 deviations as irrelevant. The following PAD histogram describes the nature of our thinking approach:

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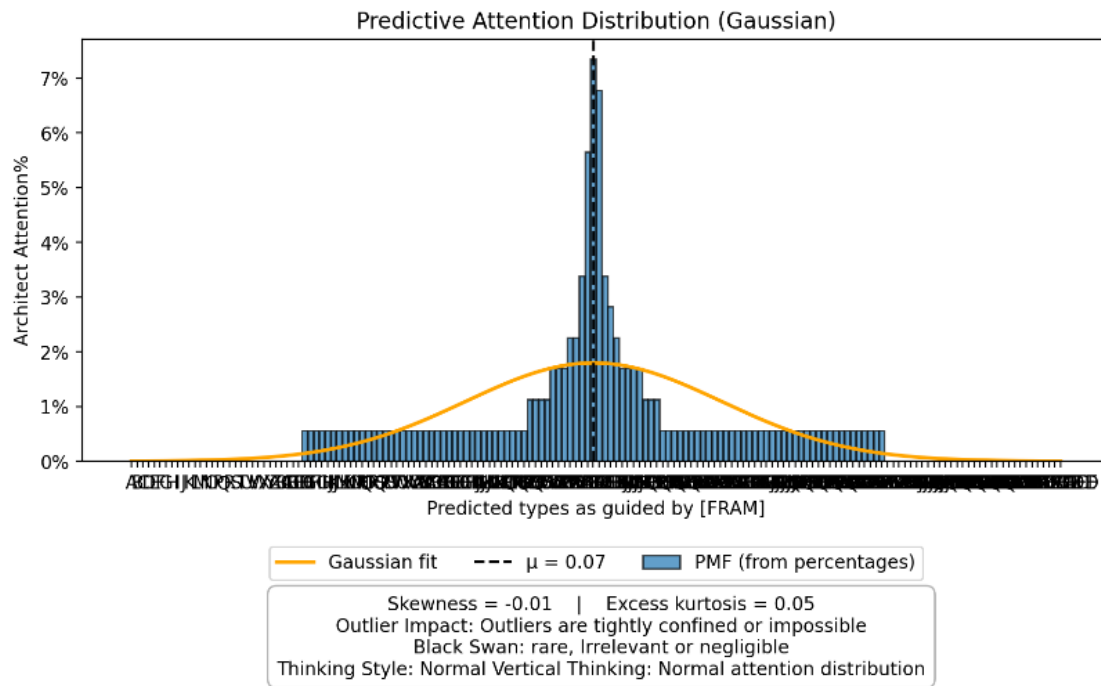


Figure G.8 FRAM predictive attention distribution analysis

The small negative kurtosis (-0.14) and skewness (0.01) indicate a tendency toward a normal distribution of attention, reinforced by the dismissal of 61 types of deviations as irrelevant. The following is the Table set of predicted situation types:

Table G.12 Predicted situations distribution as guided by FRAM approach

Predicted situation	Tally	Attention%	Originality	Typicality Test	Rational for typicality value	Novelty Test
train transit people and goods	13	7.34	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
roaming adversarial drone approach train	15	6.78	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
train traffic controller	10	5.65	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
clear train tracks zone open space facilitating safe passage for trains	6	3.39	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
monitoring tracks zone surveillance	6	3.39	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
permissible weather conditions	5	2.82	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
human operator pilot roaming adversarial drone	4	2.26	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
police force dispatch response	4	2.26	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
unsynchronised unsafe trains dispatches	7	2.26	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
environmental conditions	3	1.69	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
Police Response unit chase roaming adversarial drone	3	1.69	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
frequently delayed train	3	1.69	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
poor allocation of security monitoring resources	3	1.69	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel

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adversarial drone misses train	3	1.69	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
relay GPS / mobile network	2	1.13	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
passing through tracks zone	2	1.13	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
permissible road traffic	2	1.13	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
all trains operations	2	1.13	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
late surveillance alert	2	1.13	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
uncertain train traffic controller	2	1.13	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
impactful environmental conditions	2	1.13	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
scheduled moving train passage time	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
guided moving train driver	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
adequate fencing	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
operating tracks zone surveillance	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
dispatch train	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
alert unauthorised access	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
remotely control roaming adversarial drone	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
capable adversarial perception model	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
clear train tracks zone open space location	2	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
adversarial mission target	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
GPS signal	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
provide mobile network	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
roam clear train tracks zone open space	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
alert police force	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
dispatch police force	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
capture roaming adversarial drone	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
crossing through road traffic	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
safe transport of people	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
passenger safe arrival to destination	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
secure clear train tracks zone open space	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
police monitoring tracks zone surveillance	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
train schedule management	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
communication with moving train operators	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
miss moving train collision	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
trigger surveillance alert	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
detect roaming adversarial drone	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
wrong scheduled unsafe moving train passages	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
wrong route scheduled unsafe moving train passages	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
early guidance to unsafe moving train driver	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel

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late guidance affecting impacted all trains operations	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
wrong guidance recipient unsafe moving train driver	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
wrong direction guidance for unsafe moving train	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
inadequate tracks zone fencing	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
misplaced tracks zone fencing	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
partial coverage tracks zone fencing	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
late tracks zone surveillance	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
wrong tracks zone surveillance	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
misoriented tracks zone surveillance cameras	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
short duration tracks zone surveillance	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
early unsafe moving train dispatch	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
late unsafe moving train dispatch	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
delayed ineffective police force dispatch response	1	0.56	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
wrong unsafe moving train dispatch	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
unsafe moving train dispatched into wrong tracks	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
overextended unsafe moving train dispatch	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
early unauthorised access alert	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
late unauthorised access alert affecting impacted all trains operations	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
false unauthorised access alert	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
wrong tracks zone unauthorised access alert	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
short range unauthorised access alert	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
updated weather conditions are reported late	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
incorrect weather forecast	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
undetected adversarial drone	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
weather reports focus on irrelevant areas	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
Wrong weather conditions reported to uncertain train traffic controller	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
Wrong weather zone analysed by traffic controller	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
delayed GPS signal	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
signal data from an incorrect source or spoofed signal	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
roaming adversarial drone	1	0.56	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
adversarial drone covers wider area	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
prematurely alerted police	1	0.56	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel

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wrong police dispatch	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
insufficient police response resources	1	0.56	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
uncertain police officers approach wrong zone	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
too far dispatch location of officers	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
late chase of adversarial drone	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
excessive police chasing distance	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
road traffic related delay of police dispatch	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
late monitoring after escalation	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
false positives	1	0.56	not original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
misdirecting surveillance alert	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
too short surveillance alert	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
scheduled unsafe moving train passage time	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
unsafe dispatch train	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
Incorrect unsafe moving train information broadcasted	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
Incorrect unsafe moving train route information broadcasted	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
traveling dispatch train	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
ineffective police force dispatch response	1	0.56	original	Typical	Not outside the expected relevance in such scenario or problem space	Not novel
n/a FRAM direction deviation 12	0	negligible				
n/a FRAM direction deviation 13	0	negligible				
n/a FRAM direction deviation 14	0	negligible	0			
n/a FRAM direction deviation 15	0	negligible	0			
n/a FRAM direction deviation 16	0	negligible	0			
n/a FRAM direction deviation 19	0	negligible	0			
n/a FRAM direction deviation 20	0	negligible	0			
n/a FRAM direction deviation 21	0	negligible	0			
n/a FRAM direction deviation 22	0	negligible	0			
n/a FRAM direction deviation 23	0	negligible	0			
n/a FRAM direction deviation 25	0	negligible	0			
n/a FRAM direction deviation 26	0	negligible	0			
n/a FRAM direction deviation 27	0	negligible	0			
n/a FRAM direction deviation 28	0	negligible	0			
n/a FRAM direction deviation 32	0	negligible	0			
n/a FRAM direction deviation 34	0	negligible	0			
n/a FRAM distance deviation 2	0	negligible	0			
n/a FRAM distance deviation 4	0	negligible	0			
n/a FRAM distance deviation 12	0	negligible	0			

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n/a FRAM distance deviation 13	0	negligible	0			
n/a FRAM distance deviation 14	0	negligible	0			
n/a FRAM distance deviation 15	0	negligible	0			
n/a FRAM distance deviation 16	0	negligible	0			
n/a FRAM distance deviation 17	0	negligible	0			
n/a FRAM distance deviation 18	0	negligible	0			
n/a FRAM distance deviation 19	0	negligible	0			
n/a FRAM distance deviation 20	0	negligible	0			
n/a FRAM distance deviation 22	0	negligible	0			
n/a FRAM distance deviation 23	0	negligible	0			
n/a FRAM distance deviation 26	0	negligible	0			
n/a FRAM distance deviation 27	0	negligible	0			
n/a FRAM distance deviation 28	0	negligible	0			
n/a FRAM distance deviation 30	0	negligible	0			
n/a FRAM distance deviation 32	0	negligible	0			
n/a FRAM distance deviation 34	0	negligible	0			
n/a FRAM object deviation 12	0	negligible	0			
n/a FRAM object deviation 13	0	negligible	0			
n/a FRAM object deviation 14	0	negligible	0			
n/a FRAM object deviation 15	0	negligible	0			
n/a FRAM object deviation 16	0	negligible	0			
n/a FRAM object deviation 20	0	negligible	0			
n/a FRAM object deviation 21	0	negligible	0			
n/a FRAM object deviation 22	0	negligible	0			
n/a FRAM object deviation 25	0	negligible	0			
n/a FRAM object deviation 26	0	negligible	0			
n/a FRAM object deviation 27	0	negligible	0			
n/a FRAM object deviation 28	0	negligible	0			
n/a FRAM object deviation 32	0	negligible	0			
n/a FRAM object deviation 34	0	negligible	0			
n/a FRAM temporal deviation 8	0	negligible	0			
n/a FRAM temporal deviation 12	0	negligible	0			
n/a FRAM temporal deviation 13	0	negligible	0			
n/a FRAM temporal deviation 14	0	negligible	0			
n/a FRAM temporal deviation 15	0	negligible	0			
n/a FRAM temporal deviation 18	0	negligible	0			
n/a FRAM temporal deviation 20	0	negligible	0			

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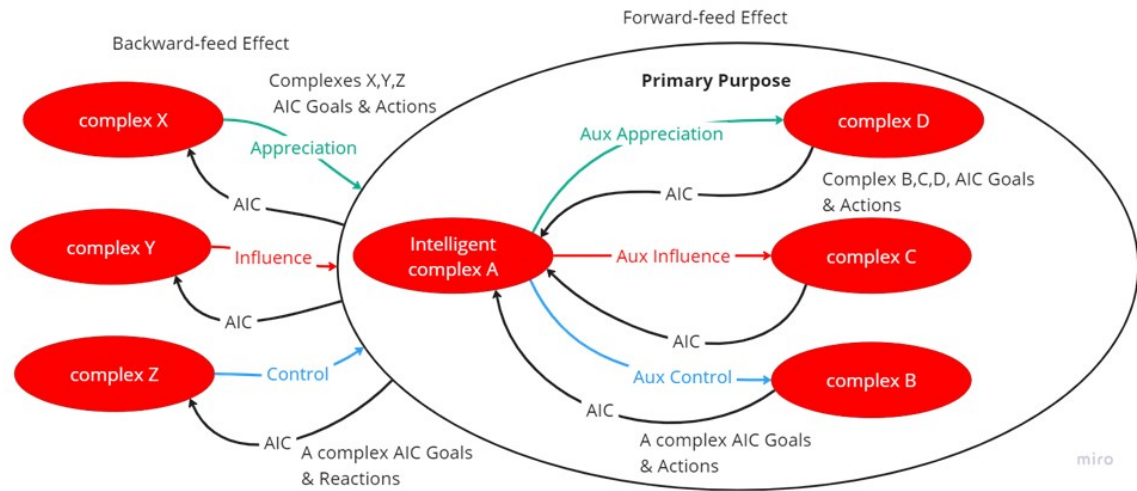
n/a FRAM temporal deviation 24	0	negligible	0			
n/a FRAM temporal deviation 27	0	negligible	0			
n/a FRAM temporal deviation 28	0	negligible	0			
n/a FRAM temporal deviation 32	0	negligible	0			
n/a FRAM temporal deviation 34	0	negligible	0			

G.4 Gap analysis of FRAM using AIC

One helpful aspect of AIC, as a predictive thinking framework, is that it enables us to evaluate the comprehensiveness of other thinking tools and to investigate the useful semantic rules they could employ to enhance the expressiveness of their models. AIC is a systems-thinking approach that aims to balance computational and lateral thinking; hence, it can be used to evaluate the general coverage of thought processes like FRAM. In this section, we also use it as an opportunity to assess how AIC can enhance the solution capability of FRAM, which we consider as part of our future research direction (see Section 9.16.2 related to the integration of AIC and FRAM).

Figure G.7 demonstrates how AIC can be semantically integrated into the FRAM ontology. In the context of AIC, Figure G.7 can be interpreted as the following:

There exists some complex A influencing complex C, in order for complex A to influence C, it must control complex B, complex E controls complex C, in order for complex correctly and reliably control B, it must complex situation D. complex A cannot influence of control complex D, complex can influence complex A ability to control E to influence C. Complex C cannot control complex A, but may be influence A through B.



AIC Framework Model

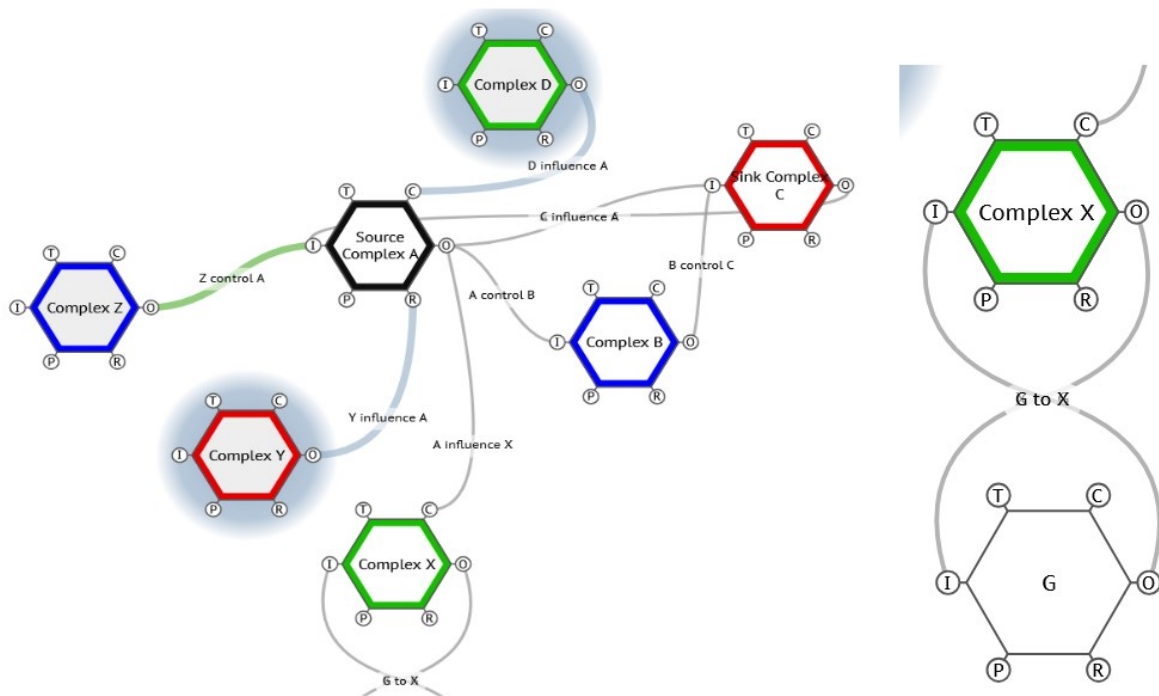


Figure G.9 modelling AIC framework using FRAM ontology. Green entity: appreciation, Red: influence, and Blue: control.

Simultaneously, there exists a complex X that appreciates A, and A influence X. Also, complex Z that controls A and Y that influence A behaviour.

When we run a model simulation cycle to visualise how AIC behaviour would emerge, we get the following behaviour:

FMI session log: 04/06/2025 15:16:18

Exit function <Sink Complex C>

Exit function <Complex X>

Entry function <Complex Z>

Interpretation Profile

Function <Complex Z>:

Output: 0

Function <Source Complex A>:

Input: All Precondition: All Resource: All Control: All Time: All

Function <Complex B>:

Input: All Precondition: All Resource: All Control: All Time: All

Function <Sink Complex C>:

Input: All

Function <Complex X>:

Input: All

Summary of FMllog

Begin initialisation

--- MODEL INITIALISATION COMPLETED.

BEGIN CYCLE 0

Background function <Complex D> has been activated

Entry function <Complex Z> has been activated

Background function <Complex Y> has been activated

BEGIN CYCLE 1

Function <Source Complex A> has been activated

BEGIN CYCLE 2

Function <Complex B> has been activated

Exit function <Complex X> has been activated

BEGIN CYCLE 3

Exit function <Sink Complex C> has been activated

The interpretation profile is set so that each function's aspects are enabled (e.g. "Input: All" means the function will activate when all its inputs are present). In each cycle, it looks for any function whose prerequisites (Inputs, Preconditions, Resources, Controls, Time) are now satisfied, and then "activates" that function, meaning it fires and produces its Output. The cycles proceed in order (0, 1, 2, ...) until every function that can run has run.

- **Cycle 0** is special: it automatically activates any **background** functions (those with no upstream dependencies) and any **entry** functions (designated "entry" by the interpretation profile).

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- In **Cycle 1** and onward, the interpreter checks which remaining functions now have all their inputs or controls available (from outputs produced in previous cycles) and activates them.
- Functions labelled **“Exit”** are typically the sinks of the model; they only activate once their single (or final) Input has arrived.

Each “has been activated” line in the log simply means: “At this cycle, all the inputs/preconditions/resources/controls needed by that function were present, so the function ran and produced its output”. The main properties of FRAM cyclic simulations:

- **Cycle 0** always runs any background or entry functions (they need no upstream inputs).
- In **Cycle n** ($n \geq 1$), any function whose inputs, preconditions, resources, controls, and time constraints are now fully met will run.
- If two or more functions become ready in the same cycle, they activate in parallel in that cycle.
- **Entry** = starts the flow (no inputs needed).
- **Background** = provides context or data (no inputs needed).
- **Exit** = final steps or sinks (only run once their one required input arrives).

Interpretation:

Cycle 0: Background + Entry functions

“Background function <Complex D> has been activated”

“Entry function <Complex Z> has been activated”

“Background function <Complex Y> has been activated”

- **Background functions** (D and Y) have no upstream dependencies in the model, they don’t need any other function’s output to run. By definition, they always fire in Cycle 0.
- **Entry function <Complex Z>** was explicitly designated in the interpretation profile with “Output: 0”. That means its output is considered available right away (no inputs needed), so the simulator fires Z immediately in Cycle 0 as well.

At the end of Cycle 0, the outputs of D, Y, and Z now exist in the system. Those outputs will serve as the “incoming conditions” that let other functions run in later cycles.

Cycle 1: Source function runs

“Function <Source Complex A> has been activated”

- In Cycle 1, the simulator checks every function that hasn’t yet run. It finds that **Source Complex A** now has all its prerequisites:
 - It needs “Input = All,” “Precondition = All,” “Resource = All,” “Control = All,” “Time = All”.
 - Because Z’s Output and the background functions’ outputs are already present (from Cycle 0), all of A’s required inputs/controls/resources are satisfied.

- Therefore, the interpreter “activates” A in Cycle 1. A’s Output is produced at the end of this cycle.

Cycle 2: Mid-chain functions and intermediate exit

“Function <Complex B> has been activated”

“Exit function <Complex X> has been activated”

- Now that A has produced its output (in Cycle 1) and the background outputs still exist, **Complex E** sees that all its required aspects are available (Input/Precondition/Resource/Control/Time = All).
 - Specifically, B’s “Input” slot is likely satisfied by A’s Output, and any other controls/resources come from Z, D, or Y.
 - So in Cycle 2, E fires and produces B’s Output.
- Also in Cycle 2, the **Exit function <Complex X>** activates.
 - An “Exit” function has no output of its own—rather, it is a designated sink that only runs once its Input is present.
 - By the time E fired, whatever E or the background functions feed into X must now be available. As soon as X sees its Input, it is eligible to run. Hence in the same cycle (Cycle 2) that E ran, X can also run if X’s single Input comes from B’s Output (or from some background).

At the end of Cycle 2, B’s Output and X’s Output (if X has any side-effect) are both in the model.

Cycle 3: Final sink completes

“Exit function <Sink Complex C> has been activated”

- In Cycle 3, the simulator looks at the remaining functions.
- **Sink Complex C** is the last function left. It only has “Input = All” listed, so it wakes up when it sees its needed Input.
 - Earlier functions must have produced the only inputs that C requires, most likely E or X (depending on how the model was coupled).
 - Since E and X both ran in Cycle 2, C now has its required Input at the start of Cycle 3.
- Therefore, in Cycle 3, C runs and finishes the mission chain. Because it’s an “Exit” function, running C means the scenario has reached its final “sink”.

Putting it all together

Note that the model did not capture the feedback loop between X and G.

1. **Initialisation** sets up the FRAM model, identifies background and entry functions, and clears any “already-run” flags.
2. Cycle 0:
 - Background functions (D, Y) run immediately because they need nothing else.

- Entry function (Z) runs immediately because its output was set to “0” (meaning “produce output at time 0”).
- 3. Cycle 1:
 - Now that the outputs from Z, D, and Y exist, the simulator checks **Source Complex A**. All of A’s inputs/controls/resources are present, so A runs.
- 4. Cycle 2:
 - With A’s output in place, **Complex E** fires next (it gets its input from A and other controls/resources from Z/D/Y).
 - Also in Cycle 2, **Exit Complex X** sees that its single input is now ready (most likely from E or a background), so it runs concurrently with B.
- 5. Cycle 3:
 - Finally, **Sink Complex C** finds that its required input—fed by E or X—is available, so it runs and completes the simulation chain.

At each cycle, the interpreter simply “wakes up” any function whose upstream dependencies were satisfied in earlier cycles. The labels **Entry** (for Z) and **Exit** (for X and C) help you see which functions kick off the process (entry) and which are the designated endpoints (exit).

G.4.1 AIC gap analysis

Some implications can lead to richer meanings and thus more room to capture complicated interactions and predict Black Swan scenarios:

- **Appreciation and Influence constraints:** AIC provide an imposed constraint on control interactions. If two complexes are in influence, then there is no direct control between them. For example, an “A influence C” interaction indicates that the control variable is disabled between A and C. As for appreciation, there is no clear aspect that can be used to link A to C(n) in an appreciative function. Stricter rules are needed to ensure that the control variable on E is disabled, preventing the Architect from making a semantic error.

The semantic meaning of AIC indicates that the designer declares the constraints are predicted to be preserved. So, when we say A influences C, it means neither A nor C shall ever be in control of the other. This means that a scenario in which one complex controls the other would constitute a Black Swan. Because the basic assumption made is “never”, so we expect control to be a very rare event as part of the design. Currently, it is not clear how FRAM capture such restrictions in behaviour.

- **No output appreciation (appreciative interaction):** The FRAM model does not allow for outputting an appreciative action between A and C(n) (the appreciated environmental complex), the same goes between X to A.

The definition of a Background function could potentially refer to an appreciated entity; however, it is restricted to being a function. This means it is not easy to model non-functional situations, such as wind, rain, or the visual appearance of an `roaming_adversarial_drone`. When we modelled the interaction between A and X, where X appreciates A but does not influence or control A, we ended up with an error in the model because it requires an input from X. So needed a background function that outputs a control or influence impact upon X. We required to have an output somewhere coming out from X, but we can not direct it to A as it is not allowed. Therefore, we needed to add a redundant entity, G, to prevent the error. However, now we have G with an output and without input, so we needed to impose a Feedback loop between X and G.

- **FRAM could not capture the nuance of unintended impact:** there is a distinct difference between A influence X and A influence C. In that C unintentionally influences X, while it is intended to influence C.
- **Delay of influence in cycles:** If we adopt a non-harmonic order of AIC, then control proceeds influence, which means that when we run the cycle, the model will activate appreciation and control interactions first, followed by influence. when we ran the model, complex in cycle 2, the interaction “A influence C” was activated before “B control C”. Influence requires “A control B” and “B control C” interactions to be activated before “A influence C”. FRAM cycles shows that:
 - Upon activation of A, E and C are controlled.
 - Then, when E controls C.

We have no means to control the sequence of activation, which means that the designer may not be able to model constrained activation delay without including two controlled functions that delay the impact of A upon C to simulate influence.

- **The simulation breaks when feedback loops are introduced:** Feedback loops are important in AIC model to capture complicatedness of systems.

When we modelled a feedback loop between G and X and then ran the simulation, we realised that the model's cyclic simulation stopped at the loop part of the model. We then tested the situation again by modelling a feedback loop between A and C, the same thing occurred. FRAM handbook mentions that hat it is **not** a flow or network model, so you do **not** draw loops as special constructs. Instead, you represent feedback by listing each function’s aspects and coupling outputs back to inputs. Any circular dependency you create is automatically treated as a loop, with no special notation beyond the standard coupling arrows. This means that the architect may miss feedback loops during modelling, thus overlooking useful requirements or solutions.

- **Risks associated with Background functions (BG):** FRAM defines a useful tool, which is BG. It defines it as “A background (BG) function is similarly a function which is assumed not to vary or to be stable during the duration of the process or activity being analysed”.

While making assumptions is useful for reducing the complexity of tasks, in compounded uncertainty problems, we need to take extra care to consider the risks we are exposing ourselves to when making them. The main risks of missing potential critical requirements are:

- BG functions are treated in isolation. Unexpected interactions between BG functions.
- The architect may forget BG functions when models scale in size.
- BG functions are assumed to be predictable alone but may exhibit unpredictable behaviour when combined.

G.4.2 Potential missed requirements when using FRAM

Below is a categorised list of general interaction types that FRAM users may overlook in analysing complicated problems, along with a brief note on how thinking in terms of AIC (Appreciation, Influence, Control) can help catch them:

1. **Missing Appreciative Interactions:** Designers may omit functions or links that merely “observe” or “sense” another entity without actively controlling it. In FRAM, it is unclear how to model a non-functional entity that impacts another entity, while the latter merely appreciates the impact of the former.

For example, function G, we needed Background functions, but the model’s restrictions make it easy to forget them altogether.

- **Why it matters:** Without an explicit “appreciation” link, you fail to record how one component simply perceives or monitors another (e.g., a drone’s computer “appreciates” wind data, but doesn’t directly control the wind).
- How AIC can help FRAM:
 - **Appreciation** reminds you to ask: “Which components simply need to know something (even if they can’t change it)?”
 - Forcing yourself to declare an appreciative interaction (e.g., “Sensor E → C(n) appreciates environmental data”) prevents accidental omission of purely observational influences.

2. Overlooked Influence Constraints (Indirect Dependencies)

- **What may be missed:** In FRAM, “A influences C” means neither A nor C can directly control one another, yet many designers may assume that any two linked functions can also “control” each other.
- **Why it matters:** If you never explicitly prohibit “A controls C” the model may allow an unrealistic coupling where A bypasses B. Omitting that constraint hides the fact that A’s effect on C must always pass-through B.
- How AIC helps:

- **Influence** forces you to distinguish between a “soft” impact ($A \rightarrow C$) versus a “hard” control ($A \rightarrow C$).
- By declaring “A influences C” (and therefore disables any direct control link), you prevent semantic mistakes, e.g., assuming A can directly manage C’s behaviour.
- You think ahead: “If I ever see A directly toggling C, that’s a Black Swan (because it violates my declared influence constraint)”.

3. Missing Unintended Interactions (Side-Effects)

- **What may be missed:** using FRAM, it’s easy to model only the **intended** chain ($A \rightarrow E \rightarrow C$) and forget that A may inadvertently influence X, or that A’s action might trigger side effects nobody planned for.

FRAM relies on functional resonance for practitioners to identify additional impacts that may have been missed during deviation analysis. However, it relies on the architect to think of that sequence of events to realise it. It may be more efficient if practitioners are explicitly prompted to consider unintended impacts during modelling.

- **Why it matters:** Some consequences arise not from a direct control or even planned influence, but from unanticipated “ripple” effects. If you never look for them, you won’t spot potential failure predictive interaction types or emergent behaviours.
- How AIC helps:
 - **Influence** again nudges you to ask: “Which functions might be loosely affected by my action, even if I didn’t intend to affect them?”
 - By labelling “C influences X (unintended),” you document that side-effect, instead of sweeping it under the rug.
 - AIC reminds you to write down both “intended” control/influence links and “unintended” actions, using a neutral sign to indicate actions that may only surface under certain conditions.

4. Practitioners may miss hidden Feedback Loops, which are sources of unpredictability

- **What’s my be missed:** FRAM’s guideline (“no special loop notation”) means you simply draw couplings; designers sometimes forget to trace a function’s Output back into the system as that function’s own Input (or another upstream Input).
- **Why it matters:** A feedback loop can dramatically change system behaviour (oscillations, homeostasis). If you never include the circular coupling (e.g., $X \rightarrow G \rightarrow X$), the model will silently break or never show the loop’s impact.
- How AIC helps:

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- **AIC** distinctions prompt you to track: “Does any function’s result eventually come back and control or influence itself (or something upstream)?”
- By forcing yourself to document “X influences G” and “G controls X,” for example, you make that loop explicit, ensuring that the tool can’t silently discard or block it.
- Even if FRAM doesn’t have a “feedback” graphic, AIC thinking makes you systematically check for any chain of dependencies that circles back.

5. Temporal Delays and Activation Order Issues

- **What may be missed:** FRAM cycles activate functions as soon as their inputs are present, but if you need a guaranteed delay (e.g., “A must wait two cycles before B”), you may forget to insert a placeholder.
- **Why it matters:** If “A influences C” should only take effect after “A controls B” and “B controls C” have both fired, but the interpreter steps them in the wrong order, you end up with an unrealistic “influence” that appears too soon.
- How AIC helps:
 - **Influence** encourages you to check: “Does an influence link really depend on prior control chains?” If so, you enforce a temporal ordering.
 - **Control** reminds you that control interactions typically happen in a strict sequence. Suppose you need to hold off on an influence until after you have control. In that case, AIC thinking pushes you to add two chained control functions (or explicit ‘delay’ functions) so that the interpreter honours the intended order.

6. Non-Functional Environmental Situations (Invisible Inputs)

- **What’s often missed:** “Wind direction”, “visual appearance of a roaming_adversarial_drone”, or “cultural norms” are not ‘active steps’, they’re context. Designers may fail to model them because FRAM expects “functions” with six aspects.
- **Why it matters:** Real-world variability often comes from these non-functional situations. If you ignore them, you’ll never capture how, for instance, a sudden change in clouds leads to unexpected sunlight reflection on object surfaces, which in turn impacts a perception model.
- How AIC helps:
 - **Appreciation** guides you to ask: “Which components need to be aware of these context situations?”

7. Neglecting Interactions Among Background Functions

- **What's often missed:** FRAM prompt the designer when specifying Background functions (BG) to assume they are stable and no need to specify CRIPTO.

In a sense, BG can be understood as a form of defining the system boundary. While they are helpful for constraining model size and helping to determine when to stop, there is a tacit risk associated with them. That is, they are assumed to be predictable and treated in isolation from one another. You might miss that one context situation actually shapes another (e.g., heavy rain degrades, and sun illumination may degrade the way objects appear in ways that were not accounted for during the training of ML models).

- **Why it matters:** If “Rain” and “Sunlight” both act as separate inputs to C(n) (environmental data) but you never connect, Sun → influences Rain → (influences) visual quality, then your training process may underestimate how the combo of those situations can indirectly degrade ML detection.
- How AIC helps:
 - **Appreciation** forces you to ask: “Which background entity appreciates another background entity?” It also permits subjective risks and scenarios to be defined relative to entities.
 - **Influence** nudges you to capture that “sunlight influences how rain influences Visual Sensor Quality”, even though neither is a “primary function”. The shape of an object may not control an autonomous system, but it can influence it through the perception component.
 - Once you draw that coupling, you see that E (navigation or ML) must account not only for raw data from D, but also for second-order variability (sunlight making the impact of rain more unpredictable).

When designers rely solely on a basic FRAM diagram, they may focus on the **straightforward control chains** (e.g., $A \rightarrow E \rightarrow C$) and may forget more subtle interactions. The AIC mindset helps reveal:

- Which entities only observe but don't control (Appreciation),
- Which entities merely exert a soft dependency but not direct control (Influence),
- Which functions have true gating or command power (Control).

By systematically asking, “Are there any influences, where no control is assumed, or appreciations I haven't drawn?” and “Am I missing a control constraint?” the architect catches nuanced dependencies, like feedback loops, unintended side-effects, shared resources, environmental situations, and rare Black Swan scenarios, before they derail the design.

G.5 Predicted Situations Distribution Analysis

To systematically evaluate the predictive attention distribution induced by different safety analysis methodologies, a data-extraction and statistical-analysis procedure was applied across

HAZOP, STAMP, and FRAM outputs. It is important to note that this process is a heuristic used to investigate how thinking approaches affect the user's predictive thinking. They are used to explore the potential of formalising predictive thinking; further validation is required to consolidate the findings, which could inform future research.

The process comprised the following steps:

G.5.1 Data Collection and Labelling Convention

For each safety methodology, predicted situations were documented at specific procedural steps, where control actions, interactions, or failure modes were defined. A structured labelling convention was adopted to ensure consistency and traceability: each situation was represented as **{adjective_noun}**, where the noun component identifies a system complex or entity, and the adjective qualifies its state or condition. For example, the situation "Drone not detected due to design or system failure" was labelled as **{track_zone_monitoring_systems}**.

The labelling convention adhered to the following principles (adhered to most cases, there were cases where the convention was slightly tailored. It is important to note that this convention is only a guideline:

1. **Label Persistence:** Once a complex (e.g., **{track_zone_monitoring_systems}**) was identified and labelled, that exact label remained unchanged throughout the analysis, regardless of how many times the same system was referenced in different contexts.
2. **State-Specific Differentiation:** If a new state or condition of the same complex was introduced, a distinct label was created by modifying the adjective component. For example, if the monitoring systems became obstructed, the situation was labelled as **{blocked_track_zone_monitoring_systems}**, representing an entirely new predicted situation rather than a repetition of the original.
3. **Situational Granularity:** Each unique label corresponds to a discrete prediction made at a particular stage of the thinking process, capturing the analyst's attention on a specific contributing situation, failure mode, or interaction at that moment.

This convention ensured that frequency counts reflected the **recurrence of attention** on specific complexes or situation types, rather than arbitrary re-labelling or semantic drift during the analysis.

G.5.2 Frequency Extraction and Tabulation

Following completion of each safety analysis, a Python script was used to extract all labelled situations from the documentation (e.g., HAZOP guideline tables, STAMP control-action deviation tables, FRAM function variability tables). The script performed the following operations:

1. **Pattern Matching:** Regular expressions identified all instances of the {adjective_noun} format within the analysis text.
2. **Frequency Counting:** Each unique label was tallied to determine how many times it appeared across the analysis, indicating the number of times that particular situation was predicted or revisited during the thinking process.
3. **Categorisation:** Labels were grouped as types of situations. Sometimes, the analysis considered the types of deviation words and then counted the number of predicted situations, meaning that the guide word had inspired the prediction of X number of situations. For example, HAZOP analysis used guide words as the predictive ontology.
4. **Tabulation and histogram generation:** The output was formatted into a frequency distribution table, listing each situation type alongside its occurrence count (Tables G.2 for HAZOP, G.6, G.7 and G.9 for STAMP and G.12 for FRAM). Then a histogram was generated.

G.5.3 Statistical Property Evaluation

The resulting frequency distribution tables were analysed statistically to characterise the **predictive attention distribution** induced by each methodology. Two key metrics were computed:

1. **Excess Kurtosis (κ):** Measures the "tailedness" of the distribution. In the context of predictive thinking, it is interpreted as the attitude of predictive thinking toward Black Swan scenarios, whether to consider them essential or negligible. Negative kurtosis (platykurtic) indicates thin tails with few or no extreme outliers; positive kurtosis (leptokurtic) indicates heavy tails with sustained attention on rare categories. Values near zero indicate mesokurtic (normal-like) distributions.
2. **Skewness (γ):** Measures asymmetry. In the context of predictive thinking, it reflects the manner of bias involved in the thinking session. Values near zero indicate symmetric distributions; large positive or negative values indicate concentration toward one end of the frequency range.

Additionally, the analysis identified **zero-frequency bins**, categories defined by the methodology's ontology (e.g., specific guidewords, deviation types), that received no predicted situations, indicating that those categories were deemed irrelevant or inapplicable.

Attention: The amount of attention expended by the architect during the analysis session. Computed by tallying the predicted situations during the analysis session, then characterise probability distribution as a ratio of No. of mentions to the total number of predicted situations. Situations that receive the least attention during the analysis session are most likely to be novel.

G.5.4 Identification of Missed Categories

For each methodology, the categories that yielded zero predicted situations were explicitly documented and counted. These "missed" categories represent **systematic epistemic gaps**: portions of the predefined analytical ontology that were not explored, either because the analyst judged them irrelevant to the scenario or because the methodology's procedural steps did not mandate their examination. For example:

- **HAZOP:** 8 guidewords (47% of the ontology) received zero attention, including "Oscillatory," "Intermittent," "Slower," and "Part of."
- **STAMP:** 49 control-action deviations (62% of theoretically possible deviations across all control actions) were marked as "not applicable" and assigned zero situations.
- **FRAM:** 61 types of deviations were considered as irrelevant, thus received 0 predicted situations.

The count of missed categories, combined with kurtosis and skewness values, provided a **distributional fingerprint** for each methodology's impact on us during analysis, enabling classification into Normal, Uniform, or Long-Tailed PAD regimes and revealing the extent to which each approach systematically biases the analyst's attention toward high-salience situations while neglecting low-salience tail categories where Black Swans may reside.

G.5.5 Python code

a) Predicted situations extraction script

```
import csv

from collections import Counter

"""
This code accepts text of any length and searches for specific characters to extract the text
located between those characters. We employ the following format to define situations:
```

{roaming_adversarial_drone}, where a situation is defined within curly braces {} and an action is enclosed within square brackets []. This script has been utilised to extract the factors and compile them into a list.

Example input text:

Too_early:_moving_train_dispatch_occurs_before_passengers_have_boarded,_causing_inconvenience_and_delays._{early_moving_train_dispatch}_Too_late:_moving_train_is_dispatched_late,_delaying_all_subsequent_schedules._{late_moving_train_dispatch}

Expected output:

['early_moving_train_dispatch', 'late_moving_train_dispatch']

And a csv file

"""

```
def extract_text_in_parentheses():
```

```
    # Prompt the user to enter a text
```

```
    input_text = input("enter your text: ")
```

```
    # List to store extracted phrases
```

```
    extracted_phrases = []
```

```
    # Temporary string to build phrases
```

```
    temp_phrase = ""
```

```
    inside_parentheses = False
```

```
    # Iterate through each character in the text
```

```
    for char in input_text:
```

```
        if char == '{': # choose the opening character
```

```
            inside_parentheses = True
```

```
            temp_phrase = ""
```

```
        elif char == '}' and inside_parentheses: # choose the closing character
```

```
            inside_parentheses = False
```

```
            # Add the phrase to the list
```

```
            extracted_phrases.append(temp_phrase)
```

```
        elif inside_parentheses:
```

```

temp_phrase += char

return extracted_phrases

predicted_factors = extract_text_in_parentheses()
print("-----")
print("-----")
print("-----")
print("-----")
print(predicted_factors)

def generate_histogram_and_csv(data_list):
    # Count the frequency of each element in the list
    frequency_count = Counter(data_list)

    # Calculate total occurrences for influence level calculation
    total_occurrences = sum(frequency_count.values())

    # Create a list of dictionaries for CSV writing
    csv_data = []
    for factor, freq in frequency_count.items():
        influence_level = freq / total_occurrences
        csv_data.append({
            "Predicted Factor": factor,
            "Frequency": freq,
            "Influence Level": influence_level
        })

    # Sort the list by frequency in descending order
    csv_data.sort(key=lambda x: x["Frequency"], reverse=True)

    # Write data to CSV file
    with open(r'C:\...\FRAM_predicted_factors.csv', 'w', newline='') as file:

```

```

writer = csv.DictWriter(file, fieldnames=["Predicted Factor", "Frequency", "Influence
Level"])

writer.writeheader()

writer.writerows(csv_data)

return r"C:\...\FRAM_predicted_factors.csv"

# Example usage with a list of elements
#example_data = ["factor1", "factor2", "factor1", "factor3", "factor2", "factor1"]
#csv_file_path = generate_histogram_and_csv(example_data)
csv_file_path = generate_histogram_and_csv(predicted_factors)

print(csv_file_path) # output the location of the CSV file that computes the frequency of factors
during analysis

```

b) Predictive Attention Distribution histogram script

```

# attention distribution analysis histogram (conditional fits + alphabetic bin labels)

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import kurtosis, skew, norm
from IPython.display import display

# ——— Excel-compatible skewness (population): SKEW.P() —————
def excel_skew_p(x):
    """
    Compute Excel's SKEW.P() (population skewness).

    Excel SKEW.P uses:
        skew_p = (1/n) * Σ [ ((xi - x̄) / s_p)^3 ]

    where:
        - n is the number of values
    """

```


- $\bar{x}$ is the mean

- s_p is the population standard deviation (ddof=0)

Returns:

float (np.nan if n == 0; 0.0 if variance is 0)

"""

```
x = np.asarray(x, dtype=float)
```

```
n = x.size
```

```
if n == 0:
```

```
    return np.nan
```

```
mean = x.mean()
```

```
s_p = x.std(ddof=0) # population standard deviation
```

```
if s_p == 0:
```

```
    return 0.0
```

```
return np.mean(((x - mean) / s_p) ** 3)
```

```
import pandas as pd
```

```
def read_table_with_fallback(path: str, sep: str):
```

"""

Try reading a delimited text file using common encodings.

Returns a DataFrame or raises the last exception if all fail.

"""

```
encodings_to_try = [
```

```
    "utf-8",
```

```
    "utf-8-sig",
```

```
    "cp1252", # Windows-1252 (very common cause of 0x92)
```

```
    "latin1", # permissive fallback
```

```
]
```

```
last_err = None
```

```

for enc in encodings_to_try:
    try:
        return pd.read_csv(path, sep=sep, encoding=enc)
    except UnicodeDecodeError as e:
        last_err = e

# If you get here, none of the encodings worked
raise last_err

# — 1) User inputs —————
path    = input("Enter path to your CSV (or TSV) file: ").strip()
sep     = input("Delimiter (',' for CSV, '\\t' for TSV) [default ',']: ").strip() or ','
factor_col = input("Enter the EXACT header of the factor column: ").strip()
freq_col = input("Enter the EXACT header of the frequency column: ").strip()
xlabel  = input("Enter the label for the x-axis: ").strip()

# — 2) Load data —————
# df = pd.read_csv(path, sep=sep)
df = read_table_with_fallback(path, sep)
for col in (factor_col, freq_col):
    if col not in df.columns:
        raise ValueError(f"Column '{col}' not found in {df.columns.tolist()}")

# — 3) Sort descending by frequency —————
df_sorted = df.sort_values(by=freq_col, ascending=False).reset_index(drop=True)

# — 4) Assign symmetric symbols: 0, +1, -1, +2, -2, ... —————
symbols = []
for i in range(len(df_sorted)):
    if i == 0:
        symbols.append(0)
    else:
        magnitude = (i + 1) // 2

```

```

    sign = 1 if (i % 2) == 1 else -1

    symbols.append(sign * magnitude)
df_sorted['Symbol'] = symbols

# — 4.1) Create alphabetical bin labels (A..Z, AA, BB, ...) —————
def _excel_like_repeated_letter_label(n: int) -> str:
    """
    1 -> A, 2 -> B, ... 26 -> Z, 27 -> AA, 28 -> BB, ... 52 -> ZZ, 53 -> AAA, ...
    Repeated-letter style per user request.
    """
    if n <= 0:
        raise ValueError("n must be >= 1")

    length = (n - 1) // 26 + 1
    within = (n - 1) % 26
    letter = chr(ord('A') + within)
    return letter * length

unique_symbols_sorted = sorted(df_sorted['Symbol'].unique())
symbol_to_alpha = {sym: _excel_like_repeated_letter_label(i + 1)
                    for i, sym in enumerate(unique_symbols_sorted)}
df_sorted['Bin'] = df_sorted['Symbol'].map(symbol_to_alpha)

# — 5) Build synthetic data array —————
values = np.repeat(df_sorted['Symbol'].values, df_sorted[freq_col].values)

# — 6) Compute statistics —————
mu = values.mean()
sigma = values.std(ddof=1)
kurt = kurtosis(values, fisher=True) # excess kurtosis

# Revert skewness to SciPy's moment skewness with bias correction
skewness_value = skew(values, bias=False)

```

```

# — 7) Determine predictive thinking classification (NEW CONDITIONS) —————

EPS = 1e-12

# Count occurrences per symbol/bin (including zeros for symbols that exist in the axis range)
min_sym, max_sym = df_sorted['Symbol'].min(), df_sorted['Symbol'].max()
all_symbols_in_range = np.arange(min_sym, max_sym + 1, 1)

# Frequency per symbol derived from the synthetic values (robust and matches plotted
# histogram bins)
counts = pd.Series(values).value_counts().reindex(all_symbols_in_range, fill_value=0)

has_zero_bin = (counts == 0).any()
all_bins_positive = (counts > 0).all()

if (-1 < kurt <= 0.5) and has_zero_bin: #and (-0.2 < skewness_value <= 0.2)
    outlier_attitude = "Outliers are tightly confined or impossible"
    black_swan_attitude = "rare, Irrelevant or negligible"
    thinking_style = "Normal Vertical Thinking: Normal attention distribution"
    fit_kind = "gaussian"
elif (kurt > 0.5): #and (abs(skewness_value) > 0.2)
    outlier_attitude = "Frequent moderate-to-extreme outliers"
    black_swan_attitude = "Rare with extreme impact"
    thinking_style = "T-Thinking (Balanced Vertical ↔ Lateral: Heavy-Tailed attention
distribution)"
    fit_kind = "heavy_tail"
elif (-1 < kurt <= 0.5) and all_bins_positive: #and (skewness_value <= 0 + EPS)
    outlier_attitude = "Outliers are not impossible, but moderately frequent"
    black_swan_attitude = "Rare with negligible impact"
    thinking_style = "Light Vertical Thinking: Light-Tail attention distribution"
    fit_kind = "light_tail"
else:
    outlier_attitude = "Everything is possible and requires equal attention"
    black_swan_attitude = "All possible situations are equally likely and impactful"
    thinking_style = "Lateral Thinking: Uniform attention distribution"
    fit_kind = "uniform"

```

```

# — 8) Plot histogram + conditional fit —————
fig, ax = plt.subplots(figsize=(10, 6))

min_sym, max_sym = df_sorted['Symbol'].min(), df_sorted['Symbol'].max()
bins = np.arange(min_sym - 0.5, max_sym + 1.5, 1)

# Histogram (COUNTS, not density)
hist_counts, hist_edges, _ = ax.hist(values, bins=bins, density=False, alpha=0.6,
edgecolor='black')

# Common x-grid for overlay lines
x = np.linspace(min_sym, max_sym, 400)

# Conditional curve fit/overlay
fit_label = ""
if fit_kind == "gaussian":
    # Scale Gaussian pdf to match histogram counts:
    # expected_count(x) ≈ N * bin_width * pdf(x)
    bin_width = hist_edges[1] - hist_edges[0]
    N = len(values)
    y_fit = norm.pdf(x, loc=mu, scale=sigma if sigma > 0 else 1.0) * N * bin_width
    ax.plot(x, y_fit, linewidth=2, label='Gaussian fit')
    fit_label = "Gaussian"
elif fit_kind == "heavy_tail":
    alpha = 1.5 + min(max(kurt, 0), 10) / 10.0 # heuristic alpha in ~[1.5, 2.5]
    c = 0.5
    y = 1.0 / (np.abs(x - mu) + c) ** alpha
    # scale to counts (area ~ N over the x-range, adjusted by bin width)
    bin_width = hist_edges[1] - hist_edges[0]
    y_fit = y / np.trapz(y, x) * len(values) * bin_width
    ax.plot(x, y_fit, linewidth=2, label=f'Heavy-tail fit (α≈{alpha:.2f})')
    fit_label = "Heavy-tail (power-law-like)"
elif fit_kind == "light_tail":

```

```

alpha = 1.5 + min(max(kurt, 0), 10) / 10.0 # heuristic alpha in ~[1.5, 2.5]
c = 0.5
y = 1.0 / (np.abs(x - mu) + c) ** alpha
bin_width = hist_edges[1] - hist_edges[0]
y_fit = y / np.trapz(y, x) * len(values) * bin_width
ax.plot(x, y_fit, linewidth=2, label=f'light-tail fit ( $\alpha \approx \{{\alpha:.2f}\}$ )')
fit_label = "Light-tail (power-law-like)"
else:
    # Uniform: constant counts per unit-x scaled by N and x-range.
    # (This is a visual reference line; it won't necessarily match discrete-bin counts perfectly.)
    width = (max_sym - min_sym) if (max_sym - min_sym) != 0 else 1.0
    bin_width = hist_edges[1] - hist_edges[0]
    y_fit = np.ones_like(x) * (len(values) / width) * bin_width
    ax.plot(x, y_fit, linewidth=2, label='Uniform fit')
    fit_label = "Uniform"

# Force integer x-ticks but display alphabetic labels
xtick_positions = np.arange(min_sym, max_sym + 1, 1)
xtick_labels = [symbol_to_alpha[pos] for pos in xtick_positions]
ax.set_xticks(xtick_positions)
ax.set_xticklabels(xtick_labels)

# Mean line
ax.axvline(mu, color='k', linestyle='--', label=f" $\mu = \{{\mu:.2f}\}$ ")

# Labels & title
ax.set_xlabel(xlabel)
ax.set_ylabel(freq_col) # now truly counts
ax.set_title(f"Predictive Attention Distribution ( $\{{\text{fit\_label}}\}$ ")

# — Move legend + metrics BELOW the x-axis (tidy, presentable) —
fig.subplots_adjust(bottom=0.35)

```

```

metrics_text = '\n'.join([
    f"σ = {sigma:.2f} | Skewness = {skewness_value:.2f} | Excess kurtosis = {kurt:.2f}",
    f"Outlier Impact: {outlier_attitude}",
    f"Black Swan: {black_swan_attitude}",
    f"Thinking Style: {thinking_style}",
])

fig.text(
    0.5, 0.06, metrics_text,
    ha='center', va='center',
    bbox=dict(boxstyle="round,pad=0.5", fc="white", ec="0.7")
)

ax.legend(
    loc='upper center',
    bbox_to_anchor=(0.5, -0.18),
    ncol=3,
    frameon=True
)

plt.show()

# — 9) Build & display mapping table —————
mapping_df = df_sorted[['Bin', factor_col, freq_col]].copy()
mapping_df.columns = ['Bin', 'Factor', 'No. predicted situations (typical saliency)']

print("\nBin ↔ Factor ↔ No. predicted situations (typical saliency) mapping:")
display(mapping_df)

```

G.5.6 Prior Studies for Originality Test

a) HAZOP output predicted situations

Data processing activities included:

- Predicted situation labels are captured between {x_y_z}

Appendix G

- Standardising predicted situation labels by alphabetically sorting the list to make sure there is no duplicated concepts. For example, “roaming_adversarial_drone”, “freely_flying_adversarial_drone” are combined under one label. Also, we made sure there is no spelling mistakes.
- Removed any spaces or new lined to make sure that no gaps might be present.
- Used the list of interactions and table G.2 for the analysis.
- The following is the raw data used for analysis.

```
{unsafe_moving_train}_pass_through_{uncertain_clear_train_tracks_zone_open_space}:_the_
unsafe_moving_train_passes_through_the_uncertain_clear_train_tracks_zone_open_space.{r
oaming_adversarial_drone}_access_{uncertain_clear_train_tracks_zone_open_space}:_the_ro
aming_adversarial_drone_accesses_the_uncertain_clear_train_tracks_zone_open_space.{unc
ertain_police_response_unit}_chase_{roaming_adversarial_drone}.{uncertain_police_respons
e}:_the_Uncertain_police_response_unit_chases_the_roaming_adversarial_drone.{uncertain_
police_response_unit}_access_{uncertain_clear_train_tracks_zone_open_space}:_the_Uncert
ain_police_response_unit_gains_access_to_the_uncertain_clear_train_tracks_zone_open_spa
ce.{roaming_adversarial_drone}_-
_{unsafe_moving_train}:_the_roaming_adversarial_drone_approaches_the_train.{uncertain_po
lice_response_unit}_deny_{unsafe_moving_train}:_the_Uncertain_police_response_unit_denie
s_unsafe_moving_train_access.
```

Table_G.2_HAZOP_process_from_problem_articulation	Interaction	Guide_word	Cause	problem_articulation_step	Consequence	problem_articulation_step	Safeguards	solution_step
Actions	solution_step	unsafe_moving_train	-	Uncertain_clear_train_tracks_zone_open_space	No	Obstruction_on_the_track_{visible_train_tracks_obstruction},_failed_equipment_{failed_equipment},_and_signalling_malfunction_{signalling_malfunction}	Frequently_delayed_train_{frequently_delayed_train},_potential_derailment_{train_derailment}_if_an_object_is_hit	Surveillance_cameras,_track_inspection_{available_ground_based_security_systems}protocols
Ensure_regular_inspection_and_provide_real-time_obstruction_alerts	unsafe_moving_train	-	Uncertain_clear_train_tracks_zone_open_space	Late	Delayed_or_advanced_scheduling_{frequently_delayed_train}	Increased_risk_of_collision_or_hazards_if_other_operations_are_unprepared_{impacted_all_trains_operations}	Automated_scheduling_and_signalling_systems{frequently_delayed_train}	Additional_scheduling_checks,_implement_manual_override_protocols_Roaming_adve

rsarial_drone_-_Uncertain_clear_train_tracks_zone_open_space No

No_deterrence_by_the_tracks_zone{uncertain_clear_train_tracks_zone_open_space}_

Potential_collision_with_a_train_{highly_probable_adversarial_drone_collision_with_unsafe_moving_train}Missing_safeguards

Increase_enforcement_measures_and_add_automated_drone_interception_mechanisms.

Roaming_adversarial_drone_-_Uncertain_clear_train_tracks_zone_open_space

More

Increased_roaming_adversarial_drone_activity_{more_roaming_adversarial_drones}_

Higher_likelihood_of_a_train-drone_collision_{highly_probable_adversarial_drone_collision_with_unsafe_moving_train},_impacted_all_trains_operations_disruption_{impacted_all_trains_operations}

Missing_safeguards

Implement_stricter_airspace_controls_and_drone_deterrence_systems_Uncertain_police_response_unit_-_roaming_adversarial_drone Faster

Faster,_inaccurate_assessment_of_the_drone's_risk_level_{roaming_adversarial_drone}

Officers_may_enter_unsafe_moving_train_zone_without_precautions_{unsafe_police_officer_movements},_leading_to_safety_risks. Missing_safeguards

Define_threat_levels_and_response_protocols_Uncertain_police_response_unit_-_roaming_adversarial_drone Other_than

Misidentification_of_the_roaming_adversarial_drone_{roaming_adversarial_drone}

Resources_misallocated,_delay_in_response_to_actual_threat_{successful_adversarial_drone_response_strategies} Missing_safeguards

Enhance_identification_accuracy_with_ai-assisted_drone_recognition_Uncertain_police_response_unit_-_uncertain_clear_train_tracks_zone_open_space As_well_as

Miscommunication_about_unsafe_moving_train_timing_or_lack_of_situational_awareness_{successful_adversarial_drone_response_strategies}

High_risk_of_collision_{train_derailment},_endangering_officer_{unsafe_police_officer_movements} Entry_control_systems,_real-time_tracking_of_officers

Implement_{alert_systems}_define_restricted_access_times_Uncertain_police_response_unit_-_uncertain_clear_train_tracks_zone_open_space Early

Communication_errors_{successful_adversarial_drone_response_strategies},_{likely_police_comms_errors}

Extended_exposure_to_potential_hazards_in_tracks_zone_{unsafe_train_tracks_zone_structure} Access_coordination,_timing_protocols

Review_and_improve_timing_protocols_for_entry_roaming_adversarial_drone_-_unsafe_moving_train Reverse Drone_control_malfunction_{roaming_adversarial_drone}

Potential_collision,_damage_to_unsafe_moving_train_{train_derailment}_infrastructure_{unsafe_train_tracks_zone_structure},_passenger_risk_{unsafe_unsecured_passengers_and_goods}	Missing_safeguards
Enable_police_or_automated_systems_to_disable/intercept_intruding_drones_roaming_adversarial_drone_-_train	Faster
Quick_speed_to_ingress_the_zone_{roaming_adversarial_drone}	
Increased_likelihood_of_collision_{train_derailment},_reduced_response_time_for_mitigation_{successful_adversarial_drone_response_strategies}	Missing_safeguards
Implement_rapid_drone_interception_measures_Uncertain_police_response_unit_-_unsafe_moving_train	No
Restricted_access,_malfunction_of_access_points_{unsafe_police_officer_movements}	
Inability_to_mitigate_threat_or_access_unsafe_moving_train_for_safety_{police_officer_capture_adversarial_drone}	Emergency_access_protocols,_clear_entry_points
Install_alternate_entry_systems,_and_enhance_emergency_accessibility_Uncertain_police_response_unit_-_train	Less
Inaccurate_timing_{likely_inaccurate_response_time},_restricted_access_{unsafe_police_officer_movements}	
Insufficient_time_to_respond_to_threats_or_ensure_area_safety_{likely_inaccurate_response_time},_Communication_systems,_access_timing_controls	
Improve_timing_and_access_protocols_to_ensure_officer_safety_	

b) STAMP output predicted situations

Data processing activities included:

- Predicted situation labels are captured between {x_y_z}
- Standardising predicted situation labels by alphabetically sorting the list to make sure there is no duplicated concepts. For example, “roaming_adversarial_drone”, “freely_flying_adversarial_drone” are combined under one label. Also, we made sure there is no spelling mistakes.
- Removed any spaces or new lined to make sure that no gaps might be present.
- Used the list of interactions and tables G.6, G.7, G.8 for the analysis.

Table_G.6_STAMP_List_of_control_actions_1.	{available_ground_based_security_systems}
detects {roaming_adversarial_drone}_2.	{available_ground_based_security_systems} alerts
{TZSC system controller}_3.	{available_ground_based_security_systems} monitors
{clear_train_tracks_zone_open_space}_4.	{roaming_adversarial_drone} triggers
{available_ground_based_security_systems}_5.	{roaming_adversarial_drone} access
{unsafe_train_tracks_zone_structure}_6.	{roaming_adversarial_drone} accesses

{clear_train_tracks_zone_open_space} (above human reach)_7. {TZSC system controller} orders {reliable_railways_traffic_control}_8. {TZSC system controller} notifies {police_force_existing_capability}_9. {TZSC_system_controller} adjusts position of {available_ground_based_security_systems} _10. {police_response_unit} dispatches {police_force_existing_capability}_11. {police_force_existing_capability} confirms attendance to {TZSC system controller}_12. {police_force_existing_capability} confirm attendance to {police_force_existing_capability}_13. {police_force_existing_capability} access {unsafe_train_tracks_zone_structure}_14. {all_trains_operations} access {unsafe_train_tracks_zone_structure}_15. {all_trains_operations} update status to {reliable_railways_traffic_control}_16. {reliable_railways_traffic_control} updates {TZSC system controller}_17. {reliable_railways_traffic_control} adjusts the movement of {all_trains_operations}_18. {reliable_railways_traffic_control} channels {unsafe_train_tracks_zone_structure}_19. {unsafe_train_tracks_zone_structure}, usher traffic for the {reliable_railways_traffic_control}_20. {unsafe_train_tracks_zone_structure} permits {all_trains_operations}_21. {unsafe_train_tracks_zone_structure} permits {police_force_existing_capability}_22. {unsafe_train_tracks_zone_structure} permits {roaming_adversarial_drone}_23. {clear_train_tracks_zone_open_space} permits {roaming_adversarial_drone}_24. {clear_train_tracks_zone_open_space} permits {available_ground_based_security_systems}__Table_G.7_STAMP_control_action_deviation_a			
analysis__Control_action	What_can_go_wrong_if_issued?		
	What_can_go_wrong_if_not_issued?	Issued_too_soon/late?	
	Issued_for_too_long/short?_1		
	False_positives_could_lead_to_unnecessary_system_responses._{false_positives},_		
	Real_drone_threats_remain_undetected,_risking_security._{false_negatives}		
	Late_detection_may_lead_to_an_accident.{likely_inaccurate_response_time} N/a_2N/a		
	Real_drone_threats_remain_undetected_{false_negatives}		
	Late_alert_may_lead_to_an_accident.{likely_inaccurate_response_time}.{late_surveillance_alert}		
	N/a_3N/a Real_drone_threats_remain_undetected_{false_negatives}_ N/a		
	Short_monitoring_intervals_may_miss_drone_re-		
	entry_{highly_probable_adversarial_drone_collision_with_unsafe_moving_train},{tracks_zone		
	_monitoring_systems_data_gaps_or_inconsistencies_4	N/a	
	Real_drone_threats_remain_undetected,_{false_negatives}_		
	Late_alert_may_lead_to_an_accident._{likely_inaccurate_response_time}{late_surveillance_alert}		
	N/a_5		
	Drones_pose_a_threat_to_ground_infrastructure_and_personnel._{unsafe_train_tracks_zone_structure}{safe_secure_police_officer}	N/a	N/a
	N/a_6		
	Drones_pose_a_threat_to_ground_infrastructure_and_personnel.{unsafe_train_tracks_z		

Appendix G

one_structure}{safe_secure_police_officer}	N/a	N/a	N/a_7N/a
Trains_continue_despite_drone_presence,_risking_collisions_or_other_accidents._{highly_probable_adversarial_drone_collision_with_unsafe_moving_train}			
late_orders_increase_accident_risk._{late_orders}	N/a_8	N/a	
Delayed_notification_leaves_the_drone_threat_unaddressed,_risking_escalation._{highly_probable_adversarial_drone_collision_with_unsafe_moving_train}			
Late_alert_the_police_may_lead_to_an_accident._{likely_inaccurate_response_time}{late_surveillance_alert}/ too soon {prematurely_alerted_police}	N/a_9		
Issued_inaccurately,_Inaccurate_Position_adjustment_can_reduce_coverage_of_critical_areas._{inaccurate_monitoring_coverage}			
Not_issued_as_needed,_{tracks_zone_monitoring_blind_spot}	N/a_	N/a_10	
Unnecessary_dispatch_strains_police_resources._{insufficient_ineffective_police_force_dispatch_response}			
Delayed_dispatch_leads_to_a_slower_response,_risking_escalation._{likely_inaccurate_response_time}{delayed_ineffective_police_force_dispatch_response}	N/a		
Brief_deployment_may_fail_to_address_threat;_prolonged_strains_resources._{uncertain_police_response_unit},_{prolonged_deployment_strains_resources}	11		
False_confirmation_could_lead_to_inadequate_management_of_the_intruding_situation._{ineffectively_managed_drone_intrusion},_{frequently_delayed_train}._			
TRAIN_TRACKS_ZONE_is_unaware_of_the_response_status,_risking_coordination._{ineffectively_managed_drone_intrusion},_{frequently_delayed_train}._			
Early_confirmation_risks_miscommunication:_late_confirmation_increases_uncertainty._{ineffectively_managed_drone_intrusion},_{frequently_delayed_train}._			
Short_confirmation_may_not_fully_verify_attendance;_long_confirmation_may_delay_other_actions._{ineffectively_managed_drone_intrusion},_{frequently_delayed_train}._	12		
False_confirmation_could_disrupt_response_timing._{ineffectively_managed_drone_intrusion},_{frequently_delayed_train}._			
Police_unit_unaware_of_actual_officer_presence,_risking_coordination_breakdown._{ineffectively_managed_drone_intrusion},_{frequently_delayed_train}._			
Early_confirmation_may_lead_to_misalignment;_late_confirmation_causes_delays_in_response._{ineffectively_managed_drone_intrusion},_{frequently_delayed_train}._			
Short_confirmation_may_cause_insufficient_information_relay,_and_long_delays._{ineffectively_managed_drone_intrusion},_{frequently_delayed_train}._	13		
Misinterpretation_could_lead_to_traffic_interference_with_{impacted_all_trains_operations},_{frequently_delayed_train},_{safe_secure_police_officer}._ {safe_secure_police_officer}			
Early_access_disrupts_{impacted_all_trains_operations};_late_risks_drone_escape._{highly_probable_adversarial_drone_collision_with_unsafe_moving_train}			

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Brief_access_may_hinder_response;_long_access_interferes_with_unsafe_moving_train_movement._{ineffectively_managed_drone_intrusion},{frequently_delayed_train}.__14	
Uncontrolled_access_risks_collision_with_personnel_or_police_in_the_zone._{highly_probable_adversarial_drone_collision_with_unsafe_moving_train},{impacted_all_trains_operations},{frequently_delayed_train},_ N/a N/a_ N/a_15 N/a	
Railways_Traffic_Controller_lacks_updated_status,_risking_{unsafe_moving_train}_safety_and_efficiency.{uncertain_train_schedule_timing}	
late_updates_reduce_rtc's_situational_awareness._{uncertain_train_schedule_timing},{reliable_railways_traffic_control} N/a_16 N/a	
TRAIN_TRACKS_ZONE_lacks_updated_status,_risking_ineffective_decision-making._{frequently_delayed_train}	
late_updates_reduce_situational_awareness._{reliable_railways_traffic_control}	
N/a_17 N/a	
Delayed_adjustments_risk_collision_or_accident._{frequently_delayed_train},{reliable_railways_traffic_control} N/a N/a_18 N/a N/a	
Late_control_may_lead_to_loss_of_income_and_penalties_to_the_{unsafe_moving_train}_operator._{lost_revenue} N/a_19 N/a	
Lack_of ushering (unclear_or_blocked_signalling)_risks_collisions_with_unauthorised_vehicles_in_the_zone._{frequently_delayed_train},{highly_probable_adversarial_drone_collision_with_unsafe_moving_train} N/a N/a_20 N/a	
Trains_are_unable_to_access_the_zones_needed_for_operations._{uncertain_train_schedule_timing},{inaccessible_tracks_zone}_{highly_probable_adversarial_drone_collision_with_unsafe_moving_train} N/a N/a_21	
Impact_on_safety_of_police_officers_{safe_secure_police_officer}	
Officers_are_unable_to_access_ground_zones_needed_for_response._{uncertain_train_schedule_timing},{inaccessible_tracks_zone}_{highly_probable_adversarial_drone_collision_with_unsafe_moving_train} N/a	
A_brief_permit_may_not_enable_full_response_or_prolonged_interference_with_operations._{ineffectively_managed_drone_intrusion},{frequently_delayed_train}.__22	
Allowing_access_would_compromise_safety_and_increase_collision_risk._{frequently_delayed_train},{inaccessible_tracks_zone}_{highly_probable_adversarial_drone_collision_with_unsafe_moving_train}N/a N/a N/a_23	
Allowing_access_would_increase_the_risk_of_collisions_with_trains_or_police_units._{frequently_delayed_train},{inaccessible_tracks_zone}_{highly_probable_adversarial_drone_collision_with_unsafe_moving_train} N/a N/a N/a_24 N/a	
Insufficient_permits_limit_airspace_monitoring_and_detection_ability._{false_negatives}	
N/a N/a_Table_G.8_STAMP_Unsafe_actions_and_safety_constraints_Unsafe_action	

Causes_problem_articulation
Mitigation_safety_constraints_solution_False_detections_clutter_the_system_with_irrelevant_data
environmental_interferences_(e.g.,_birds,_weather),_and_poor_sensor_calibration._{bad_weather} -
_implement_advanced_filtering_algorithms_to_distinguish_drones_from_false_positives._
-_regularly_calibrate_and_maintain_sensors_to_minimise_errors._ -
_machine_learning_models_are_used_to_enhance_differentiation_between_threats_and_non-threats._
Unauthorised_drones_remain_undetected_in_the_airspace
Insufficient_sensor_coverage_{false_negatives},_delayed_response_to_detection_events_{likely_inaccurate_response_time},_and_inadequate_sensitivity_of_detection_systems.
-_expand_sensor_coverage_to_eliminate_blind_spots._ -
_implement_continuous,_real-time_detection_protocols._ -
_increase_detection_sensitivity_while_minimising_false_positives_through_adaptive_sensor_calibration._
Late_monitoring_initiation_misses_early_detections.
Communication_delays_{false_negatives}-
_automate_monitoring_activation_linked_to_proximity_alerts._ -
_establish_protocols_for_immediate_monitoring_activation_upon_potential_threat_identification._
-_ensure_real-time_communication_and_minimise_human_intervention_for_quicker_response._
Short_monitoring_intervals_may_miss_drone_re-entry;_long_intervals_drain_resources
Resource_allocation_constraints,_insufficient_monitoring_capabilities,_and_incorrect_interval_configuration_{insufficient_police_response_resources}._ -
_optimize_interval_settings_based_on_historical_drone_activity_patterns._

c) FRAM output predicted situations

Data processing activities included:

- Predicted situation labels are captured between {x_y_z}
- Standardising predicted situation labels by alphabetically sorting the list to make sure there is no duplicated concepts. For example, “roaming_adversarial_drone”, “freely_flying_adversarial_drone” are combined under one label. Also, we made sure there is no spelling mistakes.
- Removed any spaces or new lines to make sure that no gaps might be present.
- Converted any entity in <>, or [] into a predicted situation: {x_y_z}
- Used the list of interactions and tables in section G3.2 for the analysis.

The following is the output list of coupled functions: 1.

{train_traffic_controller}{scheduled_moving_train_passage_time}_into_{train_transit_people_and_goods}_2.

{train_traffic_controller}{guided_moving_train_driver}_into_{train_transit_people_and_goods}_3.

{train_traffic_controller}{adequate_fencing}_into_{clear_train_tracks_zone_open_space_facilitating_safe_passage_for_trains}_4.

{train_traffic_controller}{operating_tracks_zone_surveillance}_into_{monitoring_tracks_zone_surveillance}_5.

{train_traffic_controller}{dispatch_train}_into_{train_transit_people_and_goods}_6.

{monitoring_tracks_zone_surveillance}{alert_unauthorised_access}_into_{train_traffic_controller}_7.

{human_operator_pilot_roaming_adversarial_drone}{remotely_control_roaming_adversarial_drone}_into_{roaming_adversarial_drone_approach_train}_8.

{human_operator_pilot_roaming_adversarial_drone}{capable_adversarial_perception_model}_into_{roaming_adversarial_drone_approach_train}_9.

{human_operator_pilot_roaming_adversarial_drone}{clear_train_tracks_zone_open_space_location}_into_{roaming_adversarial_drone_approach_train}_10.

{human_operator_pilot_roaming_adversarial_drone}{adversarial_mission_target}_into_{roaming_adversarial_drone_approach_train}_11.

{environmental_conditions}{permissible_weather_conditions}_into_{roaming_adversarial_drone_approach_train}_12.

{environmental_conditions}{permissible_weather_conditions}_into_{train_transit_people_and_goods}_13.

{environmental_conditions}{permissible_weather_conditions}_into_{train_traffic_controller}_14.

{relay_GPS/_mobile_network}{GPS_signal}_into_{roaming_adversarial_drone_approach_train}_15.

{relay_GPS/_mobile_network}{provide_mobile_network}_into_{roaming_adversarial_drone_approach_train}_16.

{roaming_adversarial_drone_approach_train}{clear_train_tracks_zone_open_space_location}_into_{clear_train_tracks_zone_open_space_facilitating_safe_passage_for_trains}_17.

{train_transit_people_and_goods}{passing_through_tracks_zone}_into_{clear_train_tracks_zone_open_space_facilitating_safe_passage_for_trains}_18.

{monitoring_tracks_zone_surveillance}{passing_through_tracks_zone}_into_{clear_train_tracks_zone_open_space_facilitating_safe_passage_for_trains}_19.

{train_traffic_controller}{alert_police_force}_into_{police_force_dispatch_response}_20.

{police_force_dispatch_response}{dispatch_police_force}_into_{Police_Response_unit_chase_roaming_adversarial_drone}_21.

{Police_Response_unit_chase_roaming_adversarial_drone}{capture_roaming_adversarial_drone}_into_{roaming_adversarial_drone_approach_train}_22.

{crossing_through_road_traffic}{permissible_road_traffic}_into_{police_force_dispatch_response}_23.

{clear_train_tracks_zone_open_space_facilitating_safe_passage_for_trains}{safe_transport_of_people}_into_{passenger_safe_arrival_to_destination}_24.

{Police_Response_unit_chase_roaming_adversarial_drone}{secure_clear_train_tracks_zone_open_space}_into_{clear_train_tracks_zone_open_space_facilitating_safe_passage_for_trains}_25.

{police_force_dispatch_response}{police_monitoring_tracks_zone_surveillance}_into_{monitoring_tracks_zone_surveillance}_26.

{all_trains_operations}{train_schedule_management}_into_{train_traffic_controller}_27.

{all_trains_operations}{communication_with_moving_train_operators}_into_{train_traffic_controller}_28.

{roaming_adversarial_drone_approach_train}{miss_moving_train_collision}_into_{train_transit_people_and_goods}_29.

{roaming_adversarial_drone_approach_train}{trigger_surveillance_alert}_into_{monitoring_tracks_zone_surveillance}_30.

{monitoring_tracks_zone_surveillance}{detect_roaming_adversarial_drone}_into_{roaming_adversarial_drone_approach_train}__Possible_variability_scenarios_Temporal Object

Direction

Distance_Consider_internal_or_external_variability_to_cause_the_impact_of_time_variability. Consider_internal_or_external_variability_to_cause_the_impact_of_object_variability.

Consider_internal_or_external_variability_to_cause_the_impact_of_direction_variability.

Consider_internal_or_external_variability_to_cause_the_impact_of_distance_variability.

_Too_late: the_schedule_has_been_delayed, affecting_coordination_with_other_scheduled_unsafe_moving_train_passages.{frequently_delayed_train}

Wrong_unsafe_moving_train:_incorrect_unsafe_moving_train_information_is_scheduled, impacting_other_transit_schedules.{wrong_scheduled_unsafe_moving_train_passages}

Wrong_route:_the_scheduled_time_reflects_an_incorrect_route_for_the_train, misaligning_with_traffic_flow.{wrong_route_scheduled_unsafe_moving_train_passages}

N/a_Too_early:_guidance_is_given_too_early, confusing_the_driver_about_timing.{early_guidance_to_unsafe_moving_train_driver}Too_late:_delayed_guidance_disrupts_coordinated_impacted_all_trains_operations.{late_guidance_affecting_impacted_all_trains_operations}

Wrong_unsafe_moving_train_driver:_the_guidance_is_mistakenly_sent_to_a_different_u

unsafe_moving_train_driver,_impacting_both_operations.{wrong_guidance_recipient_unsafe_moving_train_driver}

Wrong_direction:_instructions_are_provided_for_an_incorrect_track,_potentially_leading_the_unsafe_moving_train_off-course.{wrong_direction_guidance_for_unsafe_moving_train}

N/a_n/a

Inadequate_fencing:_incorrect_type_or_insufficient_fencing_materials_reduce_safety_in_the_transit_zone.{inadequate_tracks_zone_fencing}

Fencing_in_wrong_location:_fencing_is_reinforced_in_a_non-critical_area,_leaving_tracks_zones_vulnerable.{misplaced_tracks_zone_fencing}

Too_short:_adequate_fencing_only_covers_part_of_the_zone,_failing_to_protect_the_entire_transit_area.{partial_coverage_tracks_zone_fencing} Too_late:_surveillance_begins_after_the_unsafe_moving_train_is_already_in_the_zone,_potentially_missing_threats.{late_tracks_zone_surveillance}

Incorrect_zone:_surveillance_is_directed_to_a_different_zone,_leaving_the_target_area_unmonitored.{wrong_tracks_zone_surveillance}

Wrong_orientation:_surveillance_cameras_are_oriented_in_the_wrong_direction,_missing_key_areas.{misoriented_tracks_zone_surveillance_cameras}

Too_short:_surveillance_time_is_brief,_failing_to_monitor_the_train's_entire_passage.{short_duration_tracks_zone_surveillance} Too_early:_unsafe_moving_train_dispatch_occurs_before_passengers_have_boarded,_causing_inconvenience_and_delays._{early_unsafe_moving_train_dispatch} Too_late:_unsafe_moving_train_is_dispatched_late,_delaying_all_subsequent_schedules._{late_unsafe_moving_train_dispatch}{delayed_ineffective_police_force_dispatch_response}

Wrong_unsafe_moving_train_dispatched:_an_incorrect_unsafe_moving_train_is_sent,_disrupting_the_unsafe_moving_train_system_schedule._{wrong_unsafe_moving_train_dispatch}

Wrong_direction:_the_unsafe_moving_train_is_dispatched_onto_an_incorrect_track,_leading_to_misdirection._{unsafe_moving_train_dispatched_into_wrong_tracks}

Too_long:_dispatch_includes_extended_segments_beyond_its_intended_destination,_causing_inefficiency.{overextended_unsafe_moving_train_dispatch} Too_early:_alert_is_triggered_before_unauthorized_access_occurs,_leading_to_unnecessary_resource_allocation._{early_unauthorised_access_alert} Too_late:_alert_comes_after_unauthorized_access_has_already_affected_impacted_all_trains_operations.{late_unauthorised_access_alert_affecting_impacted_all_trains_operations}

False_alert:_the_system_misidentifies_authorised_personnel_as_unauthorised,_causing_operational_delays._{false_unauthorised_access_alert}

Incorrect_alert_zone:_alert_is_sent_for_an_unrelated_tracks_zone,_missing_the_target_

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area of unauthorized access. {wrong tracks zone unauthorised access alert}

Alert_range_too_short: alert fails to cover the entire tracks zone, missing parts of unauthorized access areas. {short range unauthorised access alert}	n/a	n/a	n/a
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n/a_n/a	n/a	n/a	n/a_n/a	n/a	n/a	n/a_n/a	n/a	n/a	n/a_N/a	N/a
N/a										

N/a_Too_late:_{updated_weather_conditions_are_reported_late}_after_the_unsafe_moving_train has already started transit, increasing the chance of adverse effects en route.

```
Incorrect_forecast:{incorrect_weather_forecast}_forecasts_poor_conditions,_leading_t
o_unnecessary_delays,_while_the_real_threat_(roaming_adversarial_drone)_remains_undete
cted.{undeducted_adversarial_drone}
```

Wrong_focus_area:_{weather_reports_focus_on_irrelevant_areas}_adjacent_to_the_train's path, missing critical conditions in the train's route. N/a N/a

{Wrong_weather_conditions_reported_to_uncertain_train_traffic_controller}:misreport
ed_conditions_prompt_unnecessary_delays_or_actions,_leaving_potential_threats_unmonito
red.

{Wrong_weather_zone_analysed_by_traffic_controller}:_the_controller_analysis_weather_information_for_the_wrong_zone,_misaligning_control_actions_for_the_actual_conditions_faced_by_the_train.

N/a_Too_late:_{delayed_GPS_signal}_disrupts_the_drone's_timing,_potentially_causing
it to miss or inaccurately target the train, risking broader security concerns.

[illegible]

N/a_Too_early:_{roaming_adversarial_drone},_enabling_it_to_scout_for_weak_points_and_increasing_collision_potential. N/a N/a

Too_wide:_{adversarial_drone_covers_wider_area}_roams_an_extended_area_beyond_t
he_immediate_tracks_zone,_escalating_risks_to_adjacent_impacted_all_trains_operations._T
oo_late:_unsafe_moving_train_arrival_delays_complicate_coordination_{frequently_delayed_t
rain}_and_increase_the_chance_of_collision_with_an_roaming_adversarial_drone_lingering_i
n the zone. N/a N/a

N/a_Too_early:_{prematurely_alerted_police}_before_any_threat_fully_materialises,_ris
king_resource_misallocation_and_response_fatigue._Too_late:_the_{late_surveillance_alert}_
comes_after_an_incident_has_escalated,_delaying_Uncertain_police_response_unit_and_inc
reasing_risk_to_unsafe_moving_train_and_track_safety.

Incorrect_alert_target:police_are_dispatched_to_the_wrong_area,leaving_the_true_threat_zone_unprotected. {wrong_police_dispatch}	N/a	N/a	N/a
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Wrong unit: a {insuffecient police response resources}, limiting the response capa

city_against_the_roaming_adversarial_drone.

Wrong_direction:_{uncertain_police_officers_approach_wrong_zone}_route,_reducing_their_efficiency_in_intercepting_the_drone's_path.

Too_far:_{too_far_dispatch_location_of_officers}_are_dispatched_from_too_far_station_from_the_tracks_zone_area,_wasting_resources_and_response_time._Too_late:_capture_attempt_happens_after_the_drone_completes_its_mission,_failing_to_prevent_adversarial_actions._{late_chase_of_adversarial_drone} N/a N/a

Too_far:_officers_pursue_the_drone_too_far_from_the_tracks_zone,_losing_coverage_of_other_vulnerable_areas._{excessive_police_chasing_distance}_Too_late:_traffic_control_updates_are_delayed,_causing_police_to_be_stuck_in_traffic_when_urgent_response_is_needed._{road_traffic_related_delay_of_police_dispatch} N/a N/a N/a_N/a N/a N/a

N/a_N/a N/a N/a

N/a_Too_late:_monitoring_starts_after_an_incident_has_already_escalated,_reducing_the_effectiveness_of_the_response._{late_monitoring_after_escalation}

Incorrect_surveillance_focus:_monitoring_is_directed_toward_a_less_vulnerable_zone,_leaving_critical_areas_exposed._{poor_allocation_of_security_monitoring_resources}

Wrong_orientation:_surveillance_is_focused_in_the_wrong_direction,_allowing_threats_to_go_undetected_or_poorly_addressed._{poor_allocation_of_security_monitoring_resources}

Too_short:_surveillance_monitoring_covers_only_a_portion_of_the_tracks_zone,_potentially_leaving_other_areas_unmonitored._{poor_allocation_of_security_monitoring_resources}_Too_late:_schedule_updates_are_delayed,_leading_to_unsynchronised_unsafe_trains_dispatches,_which_may_cause_delays_or_missed_connections._{unsynchronised_unsafe_trains_dispatches}

Incorrect_schedule:_an_outdated_or_incorrect_schedule_is_provided,_causing_cascading_delays_and_mismanagement_across_the_transit_system._{unsynchronised_unsafe_trains_dispatches}

Wrong_timing_focus:_scheduling_fails_to_account_for_rush_hours_or_peak_times,_leading_to_congestion_and_operational_inefficiencies._{unsynchronised_unsafe_trains_dispatches}

N/a_Too_late:_communication_delays_cause_misalignment_in_coordination,_with_potential_safety_risks_if_operators_aren't_fully_informed._{unsynchronised_unsafe_trains_dispatches}

Incorrect_communication_content:_misleading_or_incorrect_information_is_conveyed_to_the_unsafe_moving_train_operators,_risking_operational_or_safety-related_errors._{unsynchronised_unsafe_trains_dispatches}

Wrong_communication_direction:_information_is_relayed_to_the_wrong_team_or_department,_causing_unnecessary_action_or_inaction_in_key_areas._{unsynchronised_unsafe_tr

ains_dispatches}

Too_limited:_communication_fails_to_reach_all_necessary_operators,_resulting_in_fra
gmented_understanding_or_inconsistent_operational_execution..{unsynchronised_unsafe_tr
ains_dispatches}_N/a N/a N/a

N/a_Too_late:_the_alert_is_triggered_after_the_drone_has_entered_the_tracks_zone,_re
ducing_response_time_and_increasing_safety_risks..{late_surveillance_alert}

Wrong_target_alert:_the_alert_is_mistakenly_triggered_for_a_different_environmental_o
bject,_causing_distraction_and_possible_resource_misallocation..{false_positives}

Wrong_orientation:_the_alert_directs_security_focus_to_the_wrong_area,_such_as_an_
unrelated_section_of_the_tracks_zone,_delaying_interception..{misdirecting_surveillance_ale
rt}

Too_short:_the_alert_coverage_is_limited,_leaving_critical_areas_of_the_tracks_zone_u
nmonitored,_increasing_the_risk_of_undetected_unauthorised_access.{too_short_surveillanc
e_alert}_N/a N/a N/a N/a

Identify_functional_resonance_across_non-
direct_coupled_functionsTable_G.11_FRAM_definition_of_functional_resonance_Resonating_
functions_variabilities

Emergent_resonance_scenario_Negative_variability_of_{uncertain_train_traffic_controll
er}{scheduled_unsafe_moving_train_passage_time}_into_{train_transit_people_and_goods}&
uncertain_train_traffic_controller}{unsafe_dispatch_train}_into_{train_transit_people_and_goo
ds}Resonating_with:_positive_variability_of_{roaming_adversarial_drone_approach_train}{adv
ersarial_drone_misses_train}_into_{train_transit_people_and_goods}Resulting_in:Increased_t
he_likelihood_of_an_roaming_adversarial_drone_missing_a_unsafe_moving_train_collision.

Given_that:Too_late:_the_schedule_is_delayed,_impacting_the_coordination_with_othe
r_scheduled_unsafe_moving_train_passages..{frequently_delayed_train}Wrong_unsafe_movi
ng_train:_incorrect_unsafe_moving_train_information_is_scheduled,_impacting_other_transit
_schedules..{Incorrect_unsafe_moving_train_information_broadcasted}_

Wrong_route:_the_scheduled_time_reflects_an_incorrect_route_for_the_train,_misalign
ing_with_traffic_flow..{Incorrect_unsafe_moving_train_route_information_broadcasted}_Then:
Leading_to_a_change_in_unsafe_moving_train_passage_through_the_uncertain_clear_train_tr
acks_zone_open_space,_thus_reducing_the_impact_on_the_adversarial_pilot's_expected_tim
ing.__Negative_variability_of_{environmental_conditions}_output_{permissible_weather_condi
tions}_into_input_situation_of_{train_transit_people_and_goods}Resonating_with:Negative_va
riability_of_{roaming_adversarial_drone_approach_train}{adversarial_drone_misses_train}_int
o_{train_transit_people_and_goods}Resulting_in:Increased_the_likelihood_of_roaming_advers
arial_drone_unsafe_moving_train_collision.

Given_that:Incorrect_forecast:_suppose_the_forecast_anticipated_adverse_conditions;
_however,_good_conditions_emerged_earlier_than_expected._Then:This_shift_may_influence

_the_likelihood_of_an_adversarial_attack,_as_the_sudden_change_could_suggest_an_increased_probability_of_such_an_event_occurring._Negative_variability_of_{environmental_conditions}_output_{permissible_weather_conditions}_into_input_situation_of_{train_transit_people_and_goods}Resonating_with:Negative_variability_of_{traveling_dispatch_train}_output_{permissible_road_traffic}_into_control_situation_of_{ineffective_police_force_dispatch_response}Resonating_with:Negative_variability_of_{roaming_adversarial_drone_approach_train}{adversarial_drone_misses_train}_into_{train_transit_people_and_goods}Resulting_in:Increased_the_likelihood_of_roaming_adversarial_drone_unsafe_moving_train_collision.

Given_that:Incorrect_forecast:_suppose_the_forecast_anticipated_good_conditions,And_police_ignore_road_traffic_condition_since_there_is_no_bad_weather_to_cause_bad_traffic_(assuming_police_tacitly_associate_good_weather_with_easy_traffic),_forecast,_But_the_weather_changes_dramatically,_leading_to_unexpected_traffic_congestion,Too_late:_traffic_control_updates_are_delayed,_causing_police_to_be_dispatched_when_urgent_response_is_needed_during_bad_traffic.Then:This_will_lead_to_an_increase_in_the_likelihood_of_roaming_adversarial_drone_unsafe_moving_train_collisions._ _